

Folder	Paper	Claim	Tier
Readings	Folder	Gravity's asymmetry: gravity is the unique arrow-throwing, non-obviously-thermodynamic theory cleanly recoverable as an equation of state (no famous EM/Schrodinger version)	Asserted
QFT	022	Gribov ambiguity is a coarse-graining artifact (not constructively resolved)	Asserted
Black-hole	041	'Remnants NOT a dark-matter candidate'	Asserted
Black-hole	042	'Classical singularity is a coarse-graining artifact'	Asserted
Black-hole	044_5	'Leading-order match is identification, not derivation'	Asserted
Black-hole	047_5	'Four-horizon structural identification'	Asserted
Black-hole	051	'Paradox not generated' + 'information not lost'	Asserted
Entanglement	063	E-2 non-factorizability is IRREDUCIBLE	Asserted
Entanglement	071	ER=EPR-class echo	Asserted
Entanglement	072	Entanglement = unresolved individuation regime (not action-at-a-distance)	Asserted
Foundations	089/093	Wheeler-Feynman engagement: symmetric electrodynamics is mathematically consistent but substrate-operationally non-constructible in ED	Asserted
Relativistic-QM	106/107/108/111/113	Arc-R stage placements + M-verdict classification rows	Asserted
Relativistic-QM	111	'UV divergences = coarse-graining artifacts of $\omega_c \rightarrow \infty$ '; renorm = substrate-cutoff-absorption	Asserted
Constants-inherited	Constants-paper §2	$\hbar$ -as-grain-of-memory; becoming/memory/extent ($c/\hbar/l_P$); G-as-first-shadow	Asserted
Dynamics	GW-00/GW-01/Dyn-04	No graviton, no metric perturbation, no Hilbert-mode, no gauge freedom, no pseudotensor	Asserted
Substrate-evaluation	MinimalOntology	Commitment (P11) is the SOLE process primitive (ED has a spatial arena but no separate temporal one)	Asserted
Substrate-evaluation	ProxyConversionDoctrine	Proxy Conversion Doctrine D1-D4 (one rule set replacing five local patches)	Asserted
Constants-inherited	cosmology	H0 — the Hubble rate	Constant

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Constants-inherited	matter sector	Particle masses	Constant
Constants-inherited	matter sector	Gauge couplings g	Constant
Constants-inherited	matter sector	$\alpha \sim 1/137$ (fine-structure); mass ratios	Constant
Constants-inherited	Paper_012 (P-RB-1)	c — the rate of becoming (tick-to-hop conversion)	Constant
Constants-inherited	Paper_027	$G = c^3 \cdot I_P^2 / \hbar$ — Newton's constant	Constant
Constants-inherited	Paper_038_5	Λ — the cosmological constant	Constant
Constants-inherited	Paper_087	$\hbar$ — the action of one commitment (the grain of memory)	Constant
Constants-inherited	Paper_087 (P08) / 089	$I_P = I_{ED}$ — the grain (size of one hop)	Constant
Constants-inherited	Paper_029	$a_0 = c\hbar/(2\pi)$ — the MOND transition scale	Constant (scale) / POSTULATED- DISPUTED (coefficient) — 2026-09-04
QM-kinematics	009	Berry phase γ_C = holonomy of the coarse-grained P05 connection	D-via-I / Form-forced
QM-kinematics	010	Aharonov-Bohm phase from T17 bundle circulation	D-via-I / Form-forced

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QM-kinematics	011	Bloch form $\psi_k = e^{i k x} u_k(x)$ from discrete translation symmetry	D-via-I / Form-forced
QFT	022	Substrate quotient = physical-state space; BRST = coarse-grained quotient	D-via-I / Form-forced
QFT	024	Lindblad/GKSL master-equation form	D-via-I / Form-forced
Black-hole	039	Horizon = $b \rightarrow 0$ decoupling surface (Γ_{cross} collapse)	D-via-I / Form-forced
Black-hole	040	Trans-Planckian resolution (modes terminate at ω_c)	D-via-I / Form-forced
Black-hole	041	Planck-mass remnant $M^* = c \cdot l_P$ (Scenario C)	D-via-I / Form-forced
Black-hole	042	No-singularity (C_{cum} bounded, $r \geq l_P$)	D-via-I / Form-forced
Black-hole	043	Area-law FORM $S \propto A$	D-via-I / Form-forced
Black-hole	044	BHPT scattering / Regge-Wheeler form	D-via-I / Form-forced
Black-hole	045	Helicity preserved (Schwarzschild) / amplified (Kerr)	D-via-I / Form-forced
Black-hole	046	Kerr twist $g_{t\phi}$	D-via-I / Form-forced
Black-hole	047	Non-thermal Hawking correction $[1 - c_1(\omega/\omega_c)^2]$	D-via-I / Form-forced
Black-hole	047	Hawking temperature $T_H = \kappa/2\pi$ via V5 KMS	D-via-I / Form-forced
Black-hole	048	Scenario C forced (A, B excluded)	D-via-I / Form-forced
Black-hole	049	PBH relic abundance $\Omega_{\text{PBH}}^* \propto n_{\text{PBH}} \cdot M^*$	D-via-I / Form-forced
Black-hole	050	Page-curve rise + turnover + decline	D-via-I / Form-forced
Cosmology	Baryogenesis	Matter-antimatter asymmetry = structural consequence of R4 single-template admissibility + first-arrival IC	D-via-I / Form-forced
Cosmology	Cos-06	Primordial tensor modes = substrate SADDLE-HESSIAN fluctuations, SAME mechanism as post-recombination GW (GW_00)	D-via-I / Form-forced

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Gravity	GR-II	Field-equation form $G + \Lambda \cdot g = \kappa \cdot T$	D-via-I / Form-forced
QM-kinematics	012.7	Galilean adjacency/propagation bandwidth-asymmetry	Derived
Q-compute	054	UR-1.1/1.2/1.3 three unresolvedness statements ($M_{\text{eff}} < M_{\text{cap}}$; $\Gamma_{\text{cross}} > \Gamma_{\text{decoupling}}$; $\Gamma_{\text{commit}} < \Gamma_{\text{indiv}}^{-1}$)	Derived
Q-compute	054/056	Existence of Γ_{commit} , $\Gamma_{\text{indiv}}^{-1} \sim c/l_{\text{ED}}$; $M_{\text{cap}} = (\Gamma_{\text{indiv}}/\Gamma_{\text{Gamma0}})^{1/\gamma}$ as the UR-1.1 threshold	Derived
Q-compute	058b	Syndrome records are budget-free (P11 commitments)	Derived
Relativistic-QM	103	$Cl(3,1)$ frame uniqueness up to similarity	Derived
Relativistic-QM	104	Anyon prohibition in 3+1D	Derived
Substrate-evaluation	A1 (CommonCauseNotChannel)	Controlled channel capacity between distinct regions is EXACTLY zero; boundary = common-cause not channel	Derived
Substrate-evaluation	B4 (ChargeAsTopology)	Winding w in Z conserved + irreversibility-protected; orientation-blindness realizes the skeleton, withholds the field	Derived
Black-hole	BH_Thermal2Pi	Bekenstein-Hawking coefficient $1/4$ ($T = \kappa/2\pi$, $S = A/4$), GEOMETRIC route	Derived
Substrate-evaluation	CleanSubstrateVector	The parity-clean substrate is VECTOR for every channel-count ($\det H(k) \text{ even} \Rightarrow \text{winding} = 0 \text{ for all } N$); parity violation NECESSARILY spontaneous	Derived
Constants-inherited	Constants-paper §4	$G_d = c^3 \cdot l_P^{d-1} / \hbar$: the c-power is 3 in EVERY dimension (a time/mass count, not space); the spatial dimension lives entirely in l_P^{d-1}	Derived
Gravity	GR-II	ED gravity is KHRONOMETRIC (2 tensor + 1 khronon); gauge = foliation-preserving diffeos	Derived

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Gravity	GR-II	Matter conservation $\text{div } T = 0 \Leftrightarrow$ geodesic motion	Derived
Gravity	GR-II	$c_T = c$ structurally (GW170817 as identity); universal (not differential) Lorentz violation	Derived
Gravity	GR-III	Khronon scalar speed $c_s = c$	Derived
Gravity	GR-IV	$\alpha_2 = 0$ EXACTLY	Derived
Gravity	GR-IV	$\alpha_1 = -4 \cdot \lambda_{\text{local}}$; commitment FORCED sparse $\Rightarrow \lambda_{\text{local}} \ll 1$ (≥ 70 orders safe)	Derived
Gravity	MetricFromGraph	Reach law $p=1/(d-1) \Rightarrow g \sim 1/b$ in EXACTLY 3D	Derived
QFT	MS-II	Gauge group $U(N)=SU(N) \times U(1)$ from channel multiplicity	Derived
Substrate-evaluation	QuantumDarwinism	Committed records cost no V5 budget (accounting theorem) $\Rightarrow$ classical objectivity unbounded	Derived
Dark-sector	RelicLagrangian	Pressureless-dust limit $w \sim 0$ (fast-oscillating massive scalar)	Derived
Dark-sector	RelicLagrangian	Sector is TWO-COMPONENT, not one substance (L_{self} coarse-grains CANONICAL)	Derived
Dark-sector	RelicLagrangian	Relic $\rho \propto a^{-3}$ from the P11 commitment number	Derived
QM-kinematics	001	Complex polar carrier $P_K = \sqrt{b_K} \cdot e^{i \pi_K}$	Grounded
QM-kinematics	003	Born rule $f_K = c_K ^2$ as the participation-frequency limit	Grounded

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QM-kinematics	003.5	$P_K = \sqrt{b_K} \cdot e^{i \pi_K}$ amplitude carrier, complex-valued forced	Grounded
QM-kinematics	004.5	Sesquilinear inner-product form $\langle P, Q \rangle = \sum P_K^* Q_K$ (discrete)	Grounded
QM-kinematics	004.6	Continuum inner-product $\int \Psi^* \Phi \, d\mu$ on $L^2(M, d\mu)$	Grounded
QM-kinematics	005	Projective measurement (projector P_K , collapse, Born prob)	Grounded
QM-kinematics	005.5	Double-slit interference $\rho = \sum P_K ^2$; which-path = V5 injection	Grounded
QM-kinematics	006	Unitary evolution $\rightarrow$ Schrodinger $i \hbar \partial_t \Psi = H \Psi$	Grounded
QM-kinematics	006.5	Schrodinger equation via Stone's theorem	Grounded
QM-kinematics	006.6	Kinetic operator $T = p^2/2m$, the factor-2	Grounded
QM-kinematics	006.7	Schrodinger emergence in the thin-participation limit	Grounded
QM-kinematics	007	Hilbert space = Cauchy completion of the motif algebra	Grounded
QM-kinematics	008	Phase structure $S^1/U(1)$ on rule-types	Grounded
QM-kinematics	008.5	Phase-independence of bandwidth = $U(1)$ gauge invariance	Grounded
QM-kinematics	012.5	Momentum operator $p = -i \hbar \text{grad}$ = translation generator	Grounded
QM-kinematics	012.6	Heisenberg $Dx \cdot Dp \geq \hbar/2$	Grounded
QFT	013	V1 QFT-arc propagator (Kallen-Lehmann retarded form)	Grounded
QFT	014	N1+GR1: V1 finite-width + 2-point function on a curved acoustic background	Grounded

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QFT	016	Continuum gauge-covariant derivative $D_\mu = d_\mu + iA_\mu$	Grounded
QFT	017	Lorentz covariance of the continuum free-scalar equation	Grounded
QFT	017	Free scalar Klein-Gordon $(\Box + m^2)\phi = 0$ as the DCGT limit of V1	Grounded
QFT	018	Non-abelian YM potential, field strength, YM equation	Grounded
QFT	019	Continuum YM equations $D_\mu F = J$ (Euler-Lagrange)	Grounded
QFT	021	Spectral-gap FORM $\Delta \sim 1/l_*$ (Yang-Mills mass gap)	Grounded
Gravity	025	Holographic bound $N(R) \leq 4\pi R^2/l_{ED}^2$ (area law as edge-count)	Grounded
Gravity	026	Newton $a_N = GM/R^2$ from the cumulative-strain reading	Grounded
Gravity	032	Six ED-10 weak-field prerequisites fulfilled at form level	Grounded
Gravity	032.5	Free chains follow acoustic-metric geodesics	Grounded
Gravity	033	Scalar-tensor acoustic-metric field eq (covariantization)	Grounded

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Gravity	034/036	Deep-MOND $a=\sqrt{a_N a_0}$; flat-background AQUAL field eq	Grounded
Gravity	037	a_0 continuum-limit invariant (does not RG-run)	Grounded
Gravity	038.5	$\Lambda = V_1$ cosmological-scale integral over R_H ; Λ CONSTANT, $w=-1$ forced, = frozen saturation-floor	Grounded
Q-compute	053	$M_{\text{cap}} = \min(N_{\text{bw}}, N_{\text{V5}}, N_{\text{commit}})$ is one substrate object; min-structure forced by simultaneity	Grounded
Q-compute	054	The three UR-1 rate quantities have distinct PRIMARY DRIVERS (basis of A/B/C distinctness)	Grounded
Q-compute	055	A/B/C three-class EXHAUSTIVENESS — every architecture saturates one of the three constituents	Grounded
Q-compute	057	Class-B exponential gap-suppression $\Gamma_{\text{fail}} \sim \Gamma_0 e^{-\xi/\xi_0}$	Grounded
Q-compute	058	Class-C plateau $\Gamma_{\text{fail}} \rightarrow \Gamma_{\text{plateau}} > 0$ at redundancy saturation	Grounded

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Q-compute	058b	$\alpha_{\text{topological}} = 0$ (per-locus): the 058 §3.3 saturation never triggers for topological codes	Grounded
Q-compute	059	Meta-architectures (EC/DD/reservoir/hybrids) = compositions that shift which constituent binds -- NO fourth class	Grounded
Q-compute	060	M_{crit} unification: the matter-wave Q-C boundary and the qubit walls are one substrate threshold	Grounded
Entanglement	063	E-2 bilocal cross-chain term Δ_{KL} exists	Grounded
Entanglement	064	Schmidt rank $> 1 \Rightarrow$ non-factorizable	Grounded
Entanglement	065	Monogamy: total pairwise entanglement capped by the chain's V5 budget ($\sum_i E_{\{A,B_i\}} \leq W_{\text{max}}$)	Grounded
Entanglement	066	No-signaling $p(a x,y)=p(a x)$, over-determined by 3 locks	Grounded
Entanglement	068	von Neumann entropy form $S = -\text{Tr } \rho \log \rho$	Grounded
Foundations	089	Theorem N1 — V1 finite-width (G bounded, decaying, scale l_{ED} ; delta-function and inf-width limits excluded)	Grounded
Foundations	089/093	Theorem T18 — V1 strictly retarded support (the kernel-level arrow); advanced-V1 refuted by P11, symmetric-V1 non-constructible	Grounded
Foundations	090/094	V5 inherits retardation from V1 (via T18); primitive-forward-causality $\rightarrow$ phenomenological arrows inherit under DCGT	Grounded
Relativistic-QM	102	Spin-statistics $\eta = (-1)^{2s}$	Grounded
Relativistic-QM	106 (T4)	Dirac equation $(i \gamma^\mu d_\mu - mc/\hbar)\Psi = 0$	Grounded
Relativistic-QM	107	Klein-Gordon equation $(\Box + m^2 c^2/\hbar^2)\Psi = 0$	Grounded

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Relativistic-QM	108	Minimal-coupling KG + conserved 4-current	Grounded
Relativistic-QM	109	Lorentz-representation taxonomy	Grounded
Relativistic-QM	111	Primitive-level UV finiteness	Grounded
Relativistic-QM	112 (T10)	Lightlike worldlines for $\sigma=0$ rule-types	Grounded
Relativistic-QM	113	Mass structural form (Arc-M H1)	Grounded
Relativistic-QM	114 (GRH-D1)	Existence of ≥ 1 massless Case-P gauge rule-type	Grounded
Relativistic-QM	115	Vacuum + particle dual Fock structure ($\sigma=0$)	Grounded
Relativistic-QM	116	MR-R / MR-P conditional massless slot	Grounded
Substrate-evaluation	A2 (GrownNotInstalled)	A grown decoupling boundary reproduces exact channel-zero	Grounded
Substrate-evaluation	B4/Maxwell	Integral Gauss law circulation= $2 \cdot \pi \cdot w$ (loop-independent); the Maxwell action is LATENT in the coherence deficit ($\sin^2(\frac{d\phi}{2}) \sim \frac{1}{4}(\text{grad } \phi)^2$)	Grounded
State-reduction	ChannelGranularity	ED's channel/pointer basis is COARSE ($\gg I_{ED}$), so bound charges are single-channel	Grounded

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Substrate-evaluation	CleanSubstrateVector	$\gamma^5 = \text{arrow} \times \text{spontaneous spatial orientation}$; the vector clean substrate is automatically anomaly-free	Grounded
State-reduction	CollapseRate	NOT excluded by the 2020 spontaneous-radiation bound (Lindblad jump operators = channel projectors $J=g \cdot \Pi$, FORCED from P11; committed states = exact fixed points $D[\Pi]=0$; no continuous diffusion)	Grounded
State-reduction	CollapseRate	Two refinements over the rivals: a DEFINITE reduction (not Diosi's mere decoherence) + a NATIVE regulator for the point-mass divergence	Grounded
State-reduction	CollapseRate	$\tau \sim \hbar/E_G$ SCALING from ED's primitives (two branch commitment-clocks $N \sim \sqrt{b}$ the single arrow cannot hold $\rightarrow$ einselection individuates)	Grounded
Constants-inherited	Constants-paper §1	Constants ledger: $c/\hbar/l_P$ present at first events (dimensionful, $\sim$ units); $a_0/\Lambda/H_0/\text{masses}$ inherited late (dimensionless, the personality)	Grounded
Constants-inherited	Constants-paper §3	$G = c^3 \cdot l_P^2 / \hbar$; G absent from the SM; $[G_d] = m^d$ (units know the dimension count)	Grounded
Constants-inherited	Constants-paper §4	The dimension-signature in $G(l_P^{d-1})$ IS the holographic area-law / reach-law road — NOT a new independent road to 3D	Grounded
Cosmology	Cos-01	Inflation = post-SCBU substrate SATURATION regime; $a(t) = a_0 e^{\{Ht\}}$	Grounded
Cosmology	Cos-05	Λ -smallness (120-OOM) reduces to Route A + Friedmann; naive $\rho_{\text{vac}} \sim M_P^4$ STRUCTURALLY ABSENT substrate-side (no infinite mode-tower)	Grounded
Cosmology	Cos-05	Dark energy = late-universe SATURATION regime; $w=-1$ (leading order)	Grounded

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Dark-sector	DarkSector	$a_0(z) = cH(z)/(2\pi)$ EVOLUTION (flagship bet)	Grounded

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Dark-sector	DarkSector	Committed NEUTRAL RELIC candidate for clusters/CMB	Grounded
Cosmology	DF2/DF4	Low-accel galaxies CONSISTENT WITH (not counterexamples to) derived a_0 , via commitment-density saturation suppression	Grounded
Dynamics	Dyn-01	Saddle-Hessian-signature dynamics = the foundational generator; 3 regime classes (S1 collapse / S3 inflation-DE / S2 horizon-radiation)	Grounded
Dynamics	Dyn-02	Horizon-motion law: $RDE(w=1/3)$, $MDE(w=0)$, $LDE(w=-1)$ via substrate Noether stress-energy + DCGT + Friedmann	Grounded
Dynamics	Dyn-03	Radiation law: EM Larmor + GW quadrupole via Noether flux + DCGT	Grounded
Dynamics	Dyn-04	Gravitational collapse = monotonic Hessian-signature evolution toward all-compression; horizon = closed signature-class boundary	Grounded
Dynamics	Dyn-05	Compact-binary inspiral = two-saddle coupled evolution + quadrupole radiation reaction; Peters timescale	Grounded
Substrate-evaluation	GaugeStructure	Gauge group $U(N)=SU(N)\times U(1)/Z_N$ form; F^2 action; mass-gap MECHANISM $[A,A]$ lifts flat direction iff non-abelian; single $U(1)_Y$	Grounded
Substrate-evaluation	Gleason keystone	Channel orthogonality REDUCES to channel-distinctness given the representation; einselection primitive; Born non-contextuality; C selected	Grounded
Gravity	GR-I	Temporal lapse $N^2 \sim b \rightarrow$ weak-field Einstein metric (Schwarzschild branch, not Nordstrom)	Grounded
Gravity	GR-I	Emergent $g \sim 1/b$ + factor-of-two light bending	Grounded
Dynamics	GW-00	GW = propagating saddle-Hessian-signature reconfiguration (V_1 retarded); energy = local ED-gradient load (no pseudotensor)	Grounded
Dynamics	GW-00	Quadrupole source-amplitude FORM $P_{GW} \sim \langle Q\text{-triple-dot}^2 \rangle$	Grounded
Dynamics	GW-01	BH ringdown = decay of a near-horizon mixed-signature saddle; mode quantization from Hessian eigenstructure	Grounded

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Dynamics	GW-02	Stochastic background $\Omega_{GW}(f)$ = cosmologically-integrated NoetherFlux over saddle transitions	Grounded
Gravity	KM-I	Lensing tracks the MOND potential, no added vector; no new ghost; PPN-safe	Grounded
Gravity	KM-I	The khronon's deep field IS the MOND sector (unifies the two gravity programs)	Grounded
Gravity	KM-II	Orthogonality theorem: $\theta=0$ in statics $\Rightarrow$ the regulator family is invisible to every tested result	Grounded
Gravity	KM-II	Clustering dial = the soft spot (one dial sets degeneracy-cure + Cerenkov + clusters/CMB)	Grounded
Gravity	KM-II	One-Lambda thesis: $W_0 = -24 \cdot \pi^2 \cdot \Omega_{\Lambda} \sim -162$	Grounded
QFT	MS-II	Chirality γ^5 = the arrow's 'screw'	Grounded
QFT	MS-II	$\{1,2,3\} \Leftrightarrow$ internal amplitude dimension $d=3$ (Welch-bound); no stable $SU(N \geq 4)$	Grounded
QFT	MS-II	Spin-1/2 from relationality (tethered node carries $SU(2) \rightarrow SO(3)$ double cover)	Grounded
QFT	MS-II	Substrate is a $U(N)$ lattice gauge theory; escapes Nielsen-Ninomiya	Grounded
QFT	MS-II	Three spatial dimensions because the arrow must hold committed order	Grounded

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QFT	MS-II	Parity violation is non-abelian: NO parity-violating abelian force	Grounded
State-reduction	OpenPiece-i	MOND enhancement of E_G (mesoscopic masses sub- a_0 in self-gravity -> naive $\sim 5\text{-}50x$) is CLOSED by ED's own External Field Effect (Paper_029 horizon dipole sourced by total accel; Earth Newtonizes to $\sim 3.5e-6$ -> ED $\sim$ Penrose-Diosi in the lab)	Grounded
State-reduction	Positioning §3.2/§6	Locality WITHOUT superdeterminism (local commitment + A1 capacity-zero + stochastic-at-commitment; no hidden variable)	Grounded
Dark-sector	RelicLagrangian	Field content $\Phi = \sqrt{\rho/m_R} e^{i\theta} =$ coarse-grained neutral-channel participation measure	Grounded
Dark-sector	RelicLagrangian	Khronometric kinetic form L_{kin} (time-along- u + spatial c_{perp})	Grounded
Cosmology	SCBU	a_0 & $\xi_{canonical}$ = two projection MECHANISMS of one boundary $R_H = c/H_0$	Grounded
Substrate-evaluation	V5AttractiveSign/PhaseCoherence	V5's coupling IS the coherence content of the cross-chain superposition ($\sqrt{b_A b_B} \cos(\text{dpi-int } A)$); MOND cross-term constructive (P12-Coh rewards alignment)	Grounded
Substrate-evaluation	V5UnifiedBudget/QuantumDarwinism	$\rho_{loc}=1$: monogamy budget = Class-C plateau budget (one W_{max} spent two ways); per-locus live cap $N_{live} \leq W_{max}/w\text{-bar}$	Grounded
Cosmology	CCC	Planck-mass remnants dissolve at X (Scenario-C cutoff = two-chain V5 handshake; boundary emptying removes the partner)	Identification

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Cosmology	CCC	CCC's six structural elements $\leftrightarrow$ ED substrate content ($X \leftrightarrow$ SCBU; twistor $\leftrightarrow$ V5 memory; next-aeon $\leftrightarrow$ P12 re-entry; etc.)	Identification
Cosmology	CCC	Contrast-zero state unique $\implies$ CCC crossover GLUING is an identity, not a postulated match	Identification
Cosmology	Cos-02	BBN = freeze-out competition Γ_{nuc} vs substrate- $H(t)$; $Y_p \sim 0.247$	Inherited / Consilience
Cosmology	Cos-03	CMB acoustic peaks = angular projection of baryon-photon oscillator; $l_n \sim n \cdot \pi \cdot D_A / r_s$	Inherited / Consilience
Cosmology	Cos-04	Linear structure = growth of primordial perturbations $P(k, a) = D^2 T^2 P_{\text{init}}$	Inherited / Consilience
Cosmology	Cos-06	Inflationary fluctuation spectrum = sub-leading $O(\epsilon)$ perturbations on Cos-01 saturation	Inherited / Consilience
Soft-matter	P04TransportBudget_Resistivity	Resistivity saturation = Mott-Ioffe-Regel limit re-grounded in bounded P04 participation: finite per-cell bandwidth floors the mean free path at the effective grain a_{eff}	Inherited / Consilience
Foundations	096/097	$\xi_{\text{canonical}} = 1.740 \pm 0.028$ lu (cross-scale operating point), MEASURED on the certified substrate	Measured
Substrate-evaluation	AreaLaw/V5UnifiedBudget	Straddling cross-chain (V5) edge count scales as AREA (r^2), reach-independent; holographic entropy = the edge count ($\sim 0.88/\text{Planck cell}$)	Measured
Gravity	GR-III	Dynamical rule $\dot{b} = D \cdot \text{grad}^2 b - \kappa \cdot \rho$: steady state $\text{grad}^2 b \sim \rho$ (0.999); $r_s \sim M$; emergent frozen $b \rightarrow 0$ horizon; area law $S \sim A$	Measured
Gravity	GR-III	Hawking scaling $\kappa = 1/(2 r_s) \sim 1/r_h$	Measured
Black-hole	HorizonTiling	Horizon tiles ~ 1 bit/Planck cell (3 counts: holo=1, frozen 0.78, edge 0.88)	Measured
Substrate-evaluation	KineticLatticeGas/CanonicalPDE	Certified continuum = kinetic lattice-gas (BALLISTIC), NOT a diffusion PDE; and does NOT generate the UDM mobility law ($\beta \sim 0$ not 2)	Measured
Substrate-evaluation	KnotsNotCrystals/RelationaITick	No spontaneous long-range correlation-order in any sector; persistence is rhythmic (V5 binds local proper time, preserving time dilation)	Measured

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Substrate-evaluation	MassWithoutMass	Binding confines massless fronts into a sub-luminal inertial composite (mass without mass); mass(binding) != time-dilation(commit-rate)	Measured
Substrate-evaluation	Maxwell	The coherence-weighted continuum limit IS Coulomb, but the DETERMINATE substrate does NOT cast the smooth field (emergent shadow)	Measured
QFT	MS-II	Retarded ('arrow') hopping produces net chirality signatures	Measured
QFT	MS-II	Linking held in 3D, erasable in 4D	Measured
Substrate-evaluation	PhaseCoherence/V5AttractiveSign	Alignment is rewarded and reach stays FINITE (no crystal) 2D+3D; V5's attractive local-lock from P12's positive sign (flip destroys it). MEASURED ON GRAD INTENSIVELY SCORED, not on canonical Coh	Measured
Gravity	QuadraticStrain	Phase-coherence constructive sign supplied (alignment is rewarded; +Coh and -Grad both do it)	Measured
Dark-sector	RelicLagrangian	Relic is MASSIVE by V5 binding, cold-capable	Native / V5-conditional
Other-theories-advanced	(Wallstrom / Entropic Dynamics)	ED grounds the ad-hoc integer single-valuedness of Entropic Dynamics (Caticha) / resolves the Wallstrom objection	Open
QFT	019/020/021/023	YM continuum measure; OS3 derivation; mass-gap survival as $a \rightarrow 0$ (the 3 Clay-hard gaps)	Open
QFT	021/023	Confinement, asymptotic freedom, glueball spectrum, numerical Delta	Open

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Gravity	038.5	Whether Lambda frozen-floor value is FORCED or stipulated	Open
Gravity	038.5/KM-II	Ab-initio V1 boundary integral for Lambda (target $W0 = -24\pi^2 \cdot \Omega_L$); Route-A closure	Open
Black-hole	042/046	Substrate BH interior for $r < l_P$	Open
Black-hole	043	Substrate derivation of $\log g$	Open
Black-hole	045	Superradiance	Open
Black-hole	050/051	Full unitarity (constructive, not postulated)	Open
Black-hole	052_5	Substrate-anchored τ_{V5} floor (merger-lag)	Open
Q-compute	054	Formal mathematical independence of the three UR-1 rate quantities	Open
Q-compute	054	The power-law form of $\Gamma_{\text{commit}}(M_{\text{eff}})$	Open
Q-compute	056	Strategy-1 substrate-derived platform-scaling commitments (cross-platform unification)	Open
Q-compute	057	Substrate-level derivation of P-Rigidity-Gap + P-Boltz-Analogy from primitives	Open
Q-compute	058b	The repetition-code-floor discriminator (does a residual floor survive burst mitigation at cross-platform-consistent content?)	Open
Entanglement	064	derivability of P-V5-Schmidt-Generic	Open
Entanglement	068	O-vN-1: derive P-Additivity from P04 (quantitative)	Open
Entanglement	068	O-vN-2: derive P-Maximality from P11	Open
Foundations	087/088/092/093	P11 from a deeper layer (T18 $M2 \rightarrow M1$); minimality of the 13; higher-order kernels V_n ; P14 status	Open
Foundations	090	Forward derivation of V5 existence (currently a P10-licensed posit)	Open
Foundations	098/100	Substrate-level derivation of $\hbar$, c , G , a_0 ; Phase-3 Einstein-equation emergence (still SPECULATIVE)	Open

Folder	Paper	Claim	Tier
Substrate-evaluation	B4/Maxwell	Whether DCGT coarse-graining selects the Maxwell-action configuration	Open
Black-hole	BH_Thermal2Pi	Arrow-native (continuation-free) 2π	Open
Cosmology	CCC	Twistor-analog substrate integral; Big-Ring/Grand-Arc scale; cyclic vs single-aeon termination	Open
State-reduction	ChannelGranularity	Explicit channel-granularity theorem + exact distinguishability floor	Open
Substrate-evaluation	CleanSubstrateVector	Relativistic substrate->Dirac descent	Open
State-reduction	CollapseRate	Exact collapse coefficient	Open
State-reduction	CollapseRate/OpenPiece-i	Exact energy functional (from ED's quadratic-strain gravity, not Einstein-Hilbert)	Open
Cosmology	Cos-01/05	Route A (substrate-derived $I_{V5}(H_0)$ -> quantitative Λ , $N \sim 60$, H)	Open
Cosmology	Cos-02/03/04/06	Non-load-bearing inherited puzzles: 7Li , Hubble tension, S_8 , ~ 60 e-folds / termination	Open
Cosmology	Cos-03/04	CMB/clusters decoupled dark component has NO native ED mechanism	Open
Dark-sector	DarkSector	CMB spectrum computed from ED	Open
Dark-sector	DarkSector	Galactic small-scale puzzles (missing-satellite, satellite-plane)	Open
Dark-sector	DarkSector	DF2/DF4 suppression function	Open
Cosmology	DF2/DF4	Suppression function (ED analogue of MOND's ν) from primitives	Open

Folder	Paper	Claim	Tier
Dynamics	Dyn-02	OPEN-HM-Q1 (chain-class criterion rapid-V1=RDE vs slow-V1=MDE) + OPEN-HM-Q2 (stress-energy form per regime)	Open
Dynamics	Dyn-03	OPEN-RL-Q1 (source-class criterion) + OPEN-RL-Q2 (flux form per class) = THE single shared gap of the folder	Open
Substrate-evaluation	Gleason keystone	Soler lattice technical rigor (exchange-property geometry, inf-dim)	Open
QM-kinematics	Gleason-Recon (sibling)	Orthogonality REDUCED; C selected; residual = Soler	Open
Black-hole	HorizonTiling/046	Curvature emergence underneath the geometric counts	Open
Gravity	KM-I/KM-II	Clusters + CMB not discharged; khronon-standalone ABUNDANCE GAP	Open
Gravity	MetricFromGraph	Nonlinear / strong-field field equations; background-free construction	Open
QFT	MS-I/MS-II	Yang-Mills action ($\sim F^2$); electroweak/Higgs sector; EWSB; fermion masses	Open
QFT	MS-II	{1,2,3} uniqueness; weak-force specific chirality; anomaly cancellation; the linking premise	Open
QFT	MS-II	Full channel-topology $\rightarrow$ representation spectrum; spinor construction	Open
Cosmology	README	Gaussianity / CMB observed-layer debt	Open
Dark-sector	RelicLagrangian	Relic mass / free-streaming value	Open

Folder	Paper	Claim	Tier
Dark-sector	RelicLagrangian	Non-perturbative / vortex edge of L_{self}	Open
Cosmology	SCBU	$\xi_{\text{canonical}}$ substrate-derivation from H_0 (analogous to a_0)	Open
Substrate-evaluation	V5AttractiveSign/Relation alTick	Forward derivation of V5 from the primitives; the temporal envelope shape F_{V5}	Open
QM-kinematics	002	Tensor-product composition $\Psi^{AB} = \Psi^A \times \Psi^B$	Postulated
QM-kinematics	012	P-RB-1 local rate of becoming; substrate-c constancy	Postulated
QFT	015	P-NonAbelian-Analogy (non-abelian gauge structure 'by analogy')	Postulated
QFT	019	P-YM-Action-Coarse-Graining	Postulated
QFT	020	P-OS-Reflection-Positivity (OS3)	Postulated
QFT	021	P-Profile-Rescaling (YM continuum survival)	Postulated
Dark-sector	029	$a_0 = cH_0/(2\pi)$: the SCALE $\sim cH_0$ is derived; the $1/(2\pi)$ COEFFICIENT is postulated/disputed. $\sim 10\%$ match	Postulated
Gravity	030	P14 Bilocal Strain Coupling (geometric-mean)	Postulated
Gravity	035	Six acoustic-metric guardrails (C1-C6)	Postulated
Entanglement	063	E-1 factorizable $\Psi^{AB} = \Psi^A \times \Psi^B$ & the tensor-product structure	Postulated
Entanglement	064	V5-bilocal has Schmidt rank >1 generically	Postulated
Entanglement	065	Monogamy budget cap; measure saturates the budget	Postulated
Entanglement	069	ED sits at Tsirelson; PR-boxes excluded	Postulated
Foundations	087	The 13 primitives P01-P13 (the canonical postulated axiom set)	Postulated

Folder	Paper	Claim	Tier
Foundations	087	P13 = time homogeneity (not 'the tick'); the arrow = P11 + P13 + V1 retardation jointly (T18)	Postulated
Foundations	087/088	P06 (D=3+1) is extraordinarily load-bearing (underlies $4\pi R^2$ area law, inverse-square, all gravity-arc geometry)	Postulated
Foundations	087/088	P14 (Bilocal Strain Coupling) is a PLACEHOLDER, NOT in the canonical 13	Postulated
Foundations	087/088	~22 paper-specific postulates beyond the 13 (~35 total commitments)	Postulated
Foundations	094	Primitive-forward-causality (P04+P11+V1/V5 -> forward-causal at the PRIMITIVE level, not merely emergent)	Postulated
Relativistic-QM	114	P-Gauge (paper-specific postulate)	Postulated
Gravity	GR-III	$\alpha=1$ (P-Commitment-Linear) FORCED; Nordstrom excluded	Postulated
Dynamics	GW-00	P-GW-SaddleReconfiguration (GW = propagating saddle reconfiguration carried by V1)	Postulated
Gravity	QuadraticStrain	P-Quadratic-Strain (Model C)	Postulated
Substrate-evaluation	QuantumDarwinism	P-QD-LiveWeight (uncommitted branch-basis correlation consumes pairwise V5 weight)	Postulated
Cosmology	SCBU	a_0 & $\xi_{\text{canonical}}$ FORCED to share one substrate-cosmology boundary origin	Postulated
Substrate-evaluation	V5AttractiveSign/Relation aTick/MassWithoutMass	V5 existence as a cross-chain kernel	Postulated
Gravity	031	BTFR $v^4 = GM_b a_0$, slope exactly 4, zero intrinsic scatter	Prediction

Folder	Paper	Claim	Tier
Gravity	031	$a_0(z) = cH(z)/(2\pi)$ evolves factor-for-factor with H	Prediction
Gravity	038.6	Activity-dependent weak lensing $\sim \text{grad } \mu ^2$	Prediction
Black-hole	052_5	Merger-lag (existence of a delay)	Prediction
Q-compute	056	Class-A wall EXISTS as sharp + architecture-independent (the shape is the claim, not the number)	Prediction

Folder	Paper	Claim	Tier
Q-compute	057	Class-B error suppression exponential in xi across architectures	Prediction
Q-compute	058b	Class-C plateau, RE-SCOPED to broadcast-type redundancy (repetition / GHZ / cat), NOT surface-code distance	Prediction
Cosmology	Baryogenesis	Matter & antimatter follow identical V1 geodesics; no large antimatter domains	Prediction
Substrate-evaluation	CleanSubstrateVector	Parity violation is a spontaneous, contingent orientation correlated with the matter/antimatter sign	Prediction
Cosmology	Cos-05	Dark energy strictly $w=-1$, no time-variation (vs quintessence/phantom)	Prediction
Cosmology	Cos-06	Route-A4 normalization: A_s and r jointly consistent with $H_{\text{infl}} \sim \Theta^{1/2}$	Prediction
Dark-sector	DarkSector	Dark matter is WARM ($\sim 0.2-0.7$ keV), not a cold WIMP	Prediction
Dark-sector	DarkSector	Galaxies contain NO dark-matter halo at all	Prediction

Folder	Paper	Claim	Tier
Dark-sector	DarkSector	DF2/DF4 show a SHARPER external-field knee than MOND's smooth roll-off	Prediction
Dark-sector	DarkSector	No gravitational slip beyond $O(\Phi)$; activity-dependent weak-lensing enhancement in disturbed systems	Prediction
Cosmology	DF2/DF4	Group-embedded low-accel galaxies show a KNEE-like suppression sharper than MOND-EFE; void UDGs full enhancement	Prediction
Dynamics	Dyn-02	EoS confined to $\{-1, 0, 1/3\}$ + composites (no clean phantom $w < -1$, no stiff $w=1$)	Prediction
Gravity	GR-II	Scalar (breathing) GW polarization at c	Prediction
Gravity	GR-IV	α_1, α_2 preferred-frame (the keystone kill-switch)	Prediction
Dynamics	GW-00/Dyn-03/GW-01/GW-02	No non-transverse GW polarization; no GW dispersion; Larmor/Peters/ringdown/ $\Omega \sim f^{2/3}$ scaling	Prediction
Gravity	OffsetVelocity/117	Offset-velocity knee law: flat-knee-line-ceiling	Prediction

Folder	Paper	Claim	Tier
State-reduction	OpenPiece-i	Collapse rate depends on the ambient gravitational field (EFE) and is anisotropic along it	Prediction
Substrate-evaluation	QuantumDarwinism/V5UnifiedBudget	Class-C plateau <-> GHZ-width commensurate ceiling (ratios 1:1:0.88:r_QD)	Prediction
Soft-matter	UnifiedP04TransportBudget	MIR resistivity ceiling and the minimum-viscosity (KSS/Planckian) floor are ONE Adjacency-band wall -> co-onset & commensurate in strange metals	Prediction
Primitives	P01	Event-density layer existence (the foundational existence commitment; below it, no further substrate structure)	Primitive
Primitives	P02	Participation as the primitive relation (chains are the things that participate — derived composite structures)	Primitive
Primitives	P03	Channel + locus indexing; spatial homogeneity (bundles three commitments, splittable)	Primitive
Primitives	P04	Bandwidth — a non-negative additive scalar	Primitive
Primitives	P05	Polarity-transport between adjacent loci	Primitive
Primitives	P06	Spatial dimension $D = 3+1$	Primitive
Primitives	P07	Channel structure as an ontological primitive	Primitive
Primitives	P08	Substrate scale I_{ED} (= I_P empirically)	Primitive
Primitives	P09	$U(1)$ -valued polarity	Primitive
Primitives	P10	Rule-type primitive (licenses V5 as a P10 kernel rule-type)	Primitive

Folder	Paper	Claim	Tier
Primitives	P11	Commitment — irreversible, with environmental phase-randomization (THE arrow primitive)	Primitive
Primitives	P12	Stability landscape $\Sigma = \text{Coh} - \text{Str} - \text{Grad}$	Primitive
Primitives	P13	Time homogeneity; primitive event-discreteness (NOT 'the tick')	Primitive
QFT	013	V1 finite-width retarded vacuum kernel (QFT-arc presentation)	Reference
Foundations	088	Primitive Load-Bearing Audit (per-primitive dependency map + per-primitive falsifiers)	Reference
Foundations	089	V1 = the single-chain finite-width retarded vacuum kernel $K_{V1} = \theta(t-t') \cdot G(\sigma/l_{ED}^2)$	Reference
Foundations	090	V5 = the cross-chain correlation kernel (finite-memory τ_{V5} , retarded, $U(1)$ -compatible)	Reference
Foundations	095	Form-FORCED / value-INHERITED methodology; the M1/M2/M3 verdict tiers (+ label classes P/A/I/D/D-via-I/OPEN)	Reference
Foundations	095	I sub-labels (substrate-parameter-INHERITED, IC-INHERITED) + mixed-tier / component-composition rules	Reference
QM-kinematics	004.5	Tsirelson bound $2\sqrt{2}$	Selected/Inherited
QM-kinematics	012.5/012.6/006.5/006.6	$\hbar$ throughout the operator sub-arc	Selected/Inherited
QFT	015-019	Empirical gauge groups $SU(3) \times SU(2) \times U(1)$, couplings, masses, mixings	Selected/Inherited
QFT	015.5	Integer-Chern quantization C_n in $\mathbb{Z}$ of photonic Hall drift	Selected/Inherited
QFT	019/021/024	g_{YM} , mass-gap coefficient, Lindblad L_k & rates	Selected/Inherited
Gravity	027	$G = c^3 \cdot l_P^2 / \hbar$	Selected/Inherited
Gravity	029	$a_0 = c\hbar/(2\pi)$	Selected/Inherited
Gravity	038.5	Lambda magnitude $\rho_{\Lambda} \sim 1.1e-52 \text{ m}^{-2}$	Selected/Inherited

Folder	Paper	Claim	Tier
Black-hole	043	Bekenstein-Hawking $1/4$ coefficient, COMBINATORIAL route	Selected/Inherited
Black-hole	044_5	Greybody factors Γ_{ls}	Selected/Inherited
Black-hole	047_5	Universal $T = \kappa/2\pi$ for all 4 horizon types	Selected/Inherited
Black-hole	050	$t_{\text{Page}} \sim 0.54 \cdot \tau_{\text{BH}}$; c_1 , w_c , c^* values	Selected/Inherited
Q-compute	053-058	Substrate coefficients Γ_0 , γ , α , β , ξ_0 , Δ_0 , $B_{\text{cross_max}}$, $R_{\text{C_sat}}$, Γ_{plateau}	Selected/Inherited
Q-compute	056	140-250 kDa Class-A wall mass (central ~ 180 kDa)	Selected/Inherited
Q-compute	057/058	Specific codes (surface, Majorana, Shor, Steane, concatenated); rigidity/redundancy params ξ , R_{C}	Selected/Inherited
Entanglement	064	The Schmidt decomposition form itself	Selected/Inherited
Entanglement	068	$k_B / \log$ -base normalization; concrete entropy values	Selected/Inherited
Entanglement	069	Bell < Tsirelson < NS nesting; PR-box exclusion	Selected/Inherited
Entanglement	069	Tsirelson bound $2 \cdot \sqrt{2}$	Selected/Inherited
Foundations	087/089	$l_{\text{ED}} = l_{\text{P}}$ (Planck length)	Selected/Inherited
Foundations	090/089	τ_{V5} memory times (τ_{soft} =measured Maxwell time; $\tau_{\text{BH}}=l_{\text{P}}/c$; τ_{ent} system-specific); V1/V5 envelope shapes	Selected/Inherited
Relativistic-QM	110 (T8)	Canonical (anti-)commutation $[x,p]=i \hbar$	Selected/Inherited

Folder	Paper	Claim	Tier
Relativistic-QM	112 (hbar)	hbar origin = participation quantum per chain-step	Selected/Inherited
Dynamics	all	$h \sim 1/r$ amplitude falloff; $I_{\text{saddle}}/\tau_{V5}/\Theta_{ED}/H_0$ (Route A4); $r_{\text{horizon}}=2GM/c^2$, $r_{\text{ISCO}}=6GM/c^2$	Selected/Inherited
Cosmology	all Cos	H_0 value	Selected/Inherited
Cosmology	Baryogenesis	First-arrival chirality χ^* (which template = matter)	Selected/Inherited
Substrate-evaluation	CleanSubstrateVector	The casting (which force is chiral) = SU(2) fundamental pseudoreal vs SU($N \geq 3$) complex	Selected/Inherited
Cosmology	Cos-01/05/06	Uniform-Psi realizability (the saturation IC)	Selected/Inherited
Cosmology	Cos-03/04	$\Omega_c h^2 \sim 0.12$ (CMB cold-dark-matter component, $\sim 5 \times$ baryons)	Selected/Inherited
Dark-sector	DarkSector	Bullet Cluster offset (relic sails through, collisionless)	Selected/Inherited
Dynamics	Dyn-03	Larmor coefficient $1/(6 \pi \epsilon_0 c^3)$; quadrupole coefficient $1/(5c^5)$	Selected/Inherited
Substrate-evaluation	GaugeStructure/V5Unified Budget	Couplings g , scale a , charge values, EWSB; absolute V5 budget scale W_{max} ($=\Gamma_{\text{plateau}}$, t_{Page} , CKW coeff)	Selected/Inherited
Substrate-evaluation	Gleason keystone	Number field C selected from Soler's $\{R, C, H\}$	Selected/Inherited
QM-kinematics	Gleason-Recon (sibling)	Complex field C (from the Soler menu $\{R, C, H\}$)	Selected/Inherited
Dynamics	GW-00	Two transverse polarizations (h_+ , h_x); no scalar/vector/longitudinal	Selected/Inherited
Dynamics	GW-00/GW-01	GW speed $c_T = c$	Selected/Inherited
Dynamics	GW-01	QNM frequency spectrum $\omega_{\text{Imn}}(M, J) \sim c^3/GM +$ Regge-Wheeler/Teukolsky machinery	Selected/Inherited
Substrate-evaluation	ParityWall	Matter/antimatter reference (C) natively selected (first-arrival P09-phase)	Selected/Inherited

Folder	Paper	Claim	Tier
Dark-sector	RelicLagrangian	Relic mass m_R	Selected/Inherited
Dark-sector	RelicLagrangian	Relic abundance $\Omega_c \sim 0.12$	Selected/Inherited
Dark-sector	(KM-I/027-034)	Galactic dynamics from MOND/horizon, no dark-matter particle	Standing
Dark-sector	031	Baryonic Tully-Fisher slope-4 + near-zero scatter	Standing
Dark-sector	DarkSector	Cluster missing-mass shortfall filled by the relic	Standing
State-reduction	Positioning (Report §4)	Arrow selects the pointer basis (einselection PRIMITIVE); repeatability + Born non-contextuality follow	Standing
State-reduction	Positioning (Report §6)	Unitarity emergent, not fundamental (arrow forbids erasure; between-commitments = reversible QM)	Standing
State-reduction	Positioning §7	Geometry emergent $g \sim 1/b$ (both matter and geometry emergent)	Standing
Dark-sector	QuadraticStrain/030	MOND = the off-diagonal HORIZON-INTERFERENCE cross-term; $\sqrt{a_N a_0}$ forced GIVEN P-Quadratic-Strain + the bilocal-channel construction (re-scoped 2026-09-07: was 'GIVEN P14', but P14 is the postulate the quadratic-strain construction DISCHARGES, so citing it as the dependency inverted the relation). Paper_030 §5.3's regime assumption is RESTORED and is a separate, live dependency.	Standing
QM-kinematics	004	Substrate Gleason: inner-product uniqueness	Superseded
Entanglement	067	von Neumann E-6 (duplicate)	Superseded

Folder	Paper	Claim	Tier
Cosmology	Baryogenesis	P-BinaryAdmission	Superseded
Dark-sector	RelicLagrangian	L_self 'open superfluid self-term' + the khronon-mode one-substance probe	Superseded
QM-kinematics	011.5	Four QM postulates unified under the participation measure	Synthesis
QFT	023	Yang-Mills arc closes at 'Clay-relevance / structural-positive level'	Synthesis
Black-hole	047_5	Cross-class horizon identification (M3)	Synthesis
Black-hole	052	Paradox synthesis (integrates 039+047+050+041)	Synthesis
Q-compute	061	Q-Compute arc consolidation (053-060); 'decoherence-centric -> multiplicity-centric' reframing	Synthesis
Q-compute	062	Cross-domain echo: the BH horizon $\Gamma_{\text{cross}} \rightarrow 0$ collapse and Q-Compute M_{cap} saturation share one V5 mechanism	Synthesis
Entanglement	070	Bipartite architecture as one phenomenon	Synthesis
Foundations	091/092	Memory-kernel cascade $V1(\text{UV}) \rightarrow V5(\text{memory})$ under DCGT; kernel hierarchy (three-index; $V1=(I_{\text{ED}}, 1, 1)$, $V5=(I_{\text{V5}}, \text{inf}, 2)$)	Synthesis
Foundations	096/097	Cross-scale invariance (S1 kernel-content invariant); three-regime RG (UV/transition/IR)	Synthesis
Foundations	098	ED-QFT unified overview (5 sectors); Phase-1 QM at M2, DCGT M3, NS/MHD/YM/gravity M2-class	Synthesis
Foundations	099	NS / NS-MHD synthesis — NS-smoothness at Intermediate Path C / structural-positive (register identical to YM Clay-relevance); F3 = constructive blowup refutes	Synthesis
Foundations	100	Five-sector program overview (one substrate / five sectors / one methodology)	Synthesis

Folder	Paper	Claim	Tier
Substrate-evaluation	ArrowSortsTheContinuum	The arrow sorts continuum laws into three classes (structure-making native / erasing at layer-2 / preserving = edge)	Synthesis
Dynamics	Dyn-01	Cos-01 (inflation) + Cos-05 (dark energy) = the single all-expansion (S3) saddle regime at different epochs	Synthesis
Dynamics	Dyn-05/Dyn-04/GW-01/GW-02	Full inspiral->plunge->collapse->ringdown waveform closed at corpus level; BH horizon + cosmic decoupling = one substrate-parameter surface	Synthesis
Substrate-evaluation	GravityAsEquationOfState	ED grounds Jacobson's 3 assumed inputs (severance/grain/eta) and lands $G=c^3 I_P^2/\hbar$	Synthesis
Other-theories-advanced	GravityAsHorizonEquipartition	Grounds Padmanabhan's holographic equipartition (= ED's Theorem T19): ED supplies the count $N_{sur}=A/L_P^2$ (A1 severance + the measured edge-count area law)	Synthesis
Substrate-evaluation	HowPDECoarseGrainReality	Two hard shadows: YM F^2 derived-given-inputs + gap mechanism-tier (continuum OPEN); NS no-blow-up on 2 postulates; finite-grain singularity triple P04+V5+P11	Synthesis
Substrate-evaluation	HowTheoryCoarseGrainReality	Each best theory = a blur of one bandwidth field (kept->variables, discarded->symmetries); G/GR at the seam; Born un-blurred; thermo the receipt	Synthesis
Other-theories-advanced	InducedGravityArena_Sakharov	Grounds Sakharov induced gravity: ED supplies the presupposed manifold + cutoff (arena from a real substrate; grain=cutoff) + a substrate-native rigidity of gravitational strength	Synthesis
Other-theories-advanced	KhronometricEquationOfState_Jacobson	The local-Rindler $\delta Q=T-dS$ derivation for a khronometric theory: on ED's luminal family Wald entropy is pure area -> clean equilibrium -> FULL Einstein (aether among sources); shear stays exactly GR ($c_{13}=0$)	Synthesis
Substrate-evaluation	KnotsNotCrystals	ED has topological (knot/link) order, NOT crystalline	Synthesis
Substrate-evaluation	MinimalOntology	Shortest naming of ED = 'commitment and participation' over a spatial stage (floor = 3 substrate + 4 arena + 1 commitment)	Synthesis
Other-theories-advanced	MoreThanAnalogy_AnalogGravity	Grounds analog gravity's 'is there a real medium?': ED's emergent metric IS an acoustic metric (GR-I) with a REAL medium (the substrate) + the DYNAMICAL aether (the khronon) analog models lack	Synthesis

Folder	Paper	Claim	Tier
Gravity	OneField	'One field, one scale, five roles' (arrow, Einstein sector, MOND, clustering dial, Lambda-scale)	Synthesis
State-reduction	Positioning	THE arrow = gravity's time = the pointer-basis selector — one primitive, two hats (collapse variable = TIME not energy)	Synthesis
State-reduction	Positioning §3.2	ED supplies the well-defined preferred time whose absence drives Penrose's argument (collapse upstream of E_Delta)	Synthesis
Gravity	QuadraticStrain	MOND = interference cross-term: Newton=diagonal, MOND=off-diagonal of $ \sum P ^2$	Synthesis
Soft-matter	UnifiedP04TransportBudget	Unified P04 transport budget: viscosity floor & resistivity ceiling are ONE Adjacency-band wall (both = mean free path floored at a_{eff}); ED unifies MIR + Planckian dissipation + KSS viscosity bound as one substrate event	Synthesis
Substrate-evaluation	V5UnifiedBudget	Monogamy(065)+Page(050)+Class-C(058) = projections of one bounded envelope W_{max} ; R2 ratios 1:1:0.88	Synthesis
Other-theories-advanced	WhyTheCountIsArea_Susskind	Grounds the 't Hooft/Susskind holographic 'why area, not volume?': A1 severs participation wires strictly at a surface + the short-range kernel makes the straddling-edge count grow like AREA (measured $r^{2.02}$)	Synthesis

Folder	Paper	Claim	Tier
Substrate-evaluation	ArrowSortsTheContinuum	Solitons: the bare substrate does not build a reversible balance	Wall
Substrate-evaluation	GaugeStructure	Multiplicities {1,2,3} and anomalies NOT selected	Wall
Substrate-evaluation	ParityWall	The weak force's chirality (P) is NOT derived — found absent at every level checked	Wall
Substrate-evaluation	Primes (FiniteMemoryCeiling)	Finite memory saturates the prime template but PROVABLY NOT the escape (parity layer)	Wall

Entanglement	065	Monogamy in CKW form	Grounded
Foundations	097	The 0.6 three-regime-RG transition exponent	Selected/Inherited
Foundations	Paper_004_GleasonUnique	P-Gleason-Compatibility - Compatibility of the substrate bandv	Postulated
Cosmology	Paper_ED_SC_4_1_BH_Cos	P-S1-Invariant - Substrate-level S1 kernel-content is preserved	Postulated
Foundations	Paper_015_T17_GaugeFiel	P-Bundle-Definition - A substrate rule-type bundle is defined a	Postulated
Cosmology	Paper_ED_SC_4_1_BH_Cos	P-Cascade-Continuity - Coarse-graining between scale regimes	Postulated
Foundations	Paper_087_13Primitives	P-Commitment-Advancement - A commitment event that sele	Postulated
Foundations	Paper_087_13Primitives	P-Connection-Naming - Polarity-transport plays the CONNECTI	Postulated
Foundations	Paper_015_T17_GaugeFiel	P-Fiber-Naming - The channel set at substrate locus \$u\$ plays i	Postulated
Foundations	Paper_087_13Primitives	P-StructureGroup-Naming - U(1) as the substrate STRUCTURE (	Postulated
Foundations	Paper_KM-I_KhrononMON	P-Bilocal-Count - N_bilocal = N_horizon = N(R). Named 2026-0	Postulated
Foundations	Paper_024_LindbladLimit	P-Factorized-IC - Initial system-environment substrate state fac	Postulated
Foundations	Paper_UnifiedP04Transpor	P-Band-Partition - The four channel classes are DISJOINT and E	Postulated
Cosmology	Paper_080_NS_Q_Factor	P-Canonical-Operating-Point-NS-Q - The substrate-level "canor	Postulated
Soft-matter	Paper_077_NS_Smoothnes	P-Obstruction-Sufficient - The vortex-stretching obstruction (P;	Postulated
Gravity	Paper_012_RB1_RateOfBec	P-Substrate-c-Constancy - Substrate \$c\$ is invariant under any	Postulated
Gravity	Paper_033_AcousticMetric	P-AcousticMetric - (Substrate-graph acoustic metric): Substrate	Postulated
Foundations	Paper_096_CrossScaleInvar	P-Canonical-Operating-Point-ED-SC - The canonical operating p	Postulated
Foundations	Paper_078_NS_Turb_P7_C	P-Channel-Hierarchy - P07 channel structure supports a scale-i	Postulated
Foundations	Paper_027_Newtons_G	P-G-Closure - (coefficient closure on the \$G\$ assembly). The di	Postulated
Foundations	Paper_UnifiedP04Transpor	P-Locus-Bandwidth-Bound - A finite per-locus bandwidth budg	Postulated
Cosmology	Paper_085_UniversalMobil	P-Mobility-Maxwell-Correction - At finite frequency, the effecti	Postulated

Folder	Paper	Claim	Tier
Cosmology	Paper_ED_SC_4_5_SoftMa	P-Mobility-V1-Friction - Friction $\zeta \sim m/\tau_{V1}$, with finite-fr	Postulated
QFT	Paper_020_YM3_OSPositiv	P-OS-Reflection-Symmetry - The substrate is invariant under E_t	Postulated
Soft-matter	Paper_079_P4_NN_Rheolo	P-Packing-Budget-Saturation - As packing fraction ϕ appr	Postulated
Cosmology	Paper_080_NS_Q_Factor	P-Q-Factor-Form - The Q-factor at the canonical operating poir	Postulated
Soft-matter	Paper_077_NS_Smoothnes	P-R1-Sufficient-Strength - The R1 substrate-derived kinematic	Postulated
QFT	Paper_016_MinimalCouplir	P-Substrate-Covariant-Derivative-Construction - (substrate-lev	Postulated
Foundations	Paper_097_RG_Op6_Proble	P-Three-Regime-RG - The Wilsonian-RG flow of substrate effec	Postulated
QFT	Paper_013_V1_VacuumKer	P-V1-FormFactor - The momentum-space V1 form factor $\hat{\phi}$	Postulated
Cosmology	Paper_ED_SC_4_5_SoftMa	P-V5-Maxwell-Identification - V5 finite memory τ_{V5} produc	Postulated
Foundations	087	Universal free fall (equivalence-principle universality) under P1	Open
QFT	019	YM-regime applicability: the Yang-Mills results hold in the SUB	Asserted
QFT	020	Osterwalder-Schrader composite verdict: 'structural-positive-c	Asserted
Soft-matter	086.5	Photonic bandgap / negative index / cloaking reproduced from	Inherited / Consilience

Falsifier / Kill-Cond	Derived	Inherited/Open
	a real, narrow, adversarially-checked asymmetry	'why gravity is special' genuinely OPEN — best lead = 'gravity's discard IS geometry, so its EoS closes on the metric it constrains' (a lead, not a result)
		A->position
		A->position
		A->position
		A->position; greybody factors inherited
		A->position; temperature fully inherited
		A->position (audit rows 9-10); rests on P-Substrate-Unitarity; 'structural ledger, not constructive proof'
		A->assertion — 'does not supply an explicit irreducibility proof' (round-5: was D, now A)
		A->position — 'does NOT derive ER=EPR constructively'; rests on P-V5-Shared-EntanglementGravity
		A->position — interpretive framing; does not modify QM predictions
	position	empirical distinction underdetermined
	position-statement classification (per Paper_095 §3.3)	
	framework-positioning reframing, explicitly NOT a derivation	
	interpretive framing, grounded in the phase/commitment structure	
		A->position (four substrate-ontology replacement claims)
		A P11-pivoted audit (pivot disclosed); 'keystone synthesis' is an account, not a theorem; 3D selected not derived
		methodological, falsifiable structure; the alpha-topological encoding-dependence was created then SETTLED same week (alpha_topological=0, Paper_058b)
		INHERITED (empirical cosmological content)

Falsifier / Kill-Cond	Derived	Inherited/Open
		INHERITED (bound b; the matter sector, 'the unfinished room')
		INHERITED
		INHERITED (the physics-defining dimensionless numbers = the universe's late-acquired personality)
		EVENT-SCALE (present at the first events; dimensionful; = 1 in natural units — a unit, not a number)
	a COMBINATION, not an independent input (adds no new information); ABSENT from the entire Standard Model	DERIVED-combination; the 'first shadow' at the mass<->geometry seam; least-fundamental yet first-emergent
		INHERITED / late (value via H_0 ; $w=-1$ forced; smallness open)
		EVENT-SCALE (present at the first events; = 1 natural units)
		EVENT-SCALE; $I_{ED}=I_P$ set empirically via Newton-recovery (Paper_027 §5.3)
		INHERITED / late (needs a Hubble horizon; so late it EVOLVES, $a_0(z)$)
	D-via-I (holonomy on Paper_016 connection)	numerical γ_C inherited; non-abelian/off-diagonal OPEN
	D-via-I (fiber-bundle holonomy)	$e/\hbar$ inherited; paper: 'ED does NOT newly predict AB'

Falsifier / Kill-Cond	Derived	Inherited/Open
	D-via-I (abelian rep theory)	band content inherited
	D-via-I: fiber-bundle quotient + DCGT	constructive Gribov resolution OPEN
	FORM-FORCED as the Markovian limit of V5-modulated dynamics (D-via-I)	specific L_k , rates I ; non-Markovian OPEN
	D mechanism (decoupling-surface)	r_H location + κ INHERITED from GR (§3.2 honest-ack: r_H a CG reference, not derived)
	D-via-I	conditional on P-V5-UV-Cutoff + P-Cutoff-Saturation
	D-via-I	conditional on P-Cutoff-Saturation-Endpoint + P-Remnant-Stability + 048; c^* inherited
	D-via-I	conditional on P-Bandwidth-Boundedness + P-Substrate-Interior-Cutoff; interior OPEN
	D-via-I (form forced)	conditional on P-Horizon-Participation + P-Multiplicity-g; coefficient inherited by this route
	form-forced	conditional on P-Horizon-Ingoing-Only + P-Asymptotic-Outgoing; cross-sections inherited
	form-forced	conditional on P-Helicity-Channel + P-Kerr-Helicity-Coupling; superradiance OPEN
	form-forced	conditional on P-Substrate-Vorticity + P-Twist-Coarse-Graining; a inherited; interior OPEN
	D form	conditional on P-V5-Even; w_c chosen-to-match, c_1 inherited
	KMS => thermal = D	κ I ; Euclidean/conical 2π = I-via-DCGT (audit rows 5-8, not substrate-derived)
	D-via-I	conditional on P-Resummation-Convergence + P-Endpoint-Truncation; coeffs inherited
	D-via-I	conditional on P-One-Remnant-Per-PBH
	D	conditional on P-V5-EntBudget + P-Re-routing; $t_{Page} \sim 0.54 \cdot \tau_{BH}$ inherited; unitarity OPEN
	single-template admission (coherence functional + V5 phase-coherence + no-slack, D-via-I); global lock-in	$\eta_B \sim 6e-10$ I (Planck/BBN); WHICH chirality wins = IC-INHERITED
	the cross-arc identification (D-via-I); 'cleanest cross-arc unification in the arc'	consistency relation $n_T = -r/8$ inherited

Falsifier / Kill-Cond	Derived	Inherited/Open
	Lovelock, given metric + conserved source + 2nd-order	forced IFF purely metric — the open condition, which §5 FAILS (ED is khronometric, not pure GR)
	asymmetry FORCED by P03 (adjacency invariant) + P04 (propagation transforms)	full Galilean/Bargmann cohomology OPEN (P03 necessary, not sufficient); m inherited
	algebraic from Gamma_commit + Gamma_indiv (given A->regime power-law); UR-1.2 from V5 (Paper_090)	rests on the A->regime power-law form; coefficients inherited
	existence from P11+P12+V1 finite width; M_cap from Gamma_commit(M_cap) =Gamma_indiv	numerical Gamma0, gamma inherited (empirical anchoring)
	from the QuantumDarwinism record-bandwidth accounting theorem	
	unique irreducible 4-dim rep of Cl(3,1); all gamma-realizations similarity-equivalent (Pauli + Schur)	basis (Dirac/Weyl/Majorana) = choice of S, not value inheritance; conditional on Paper_017 signature
an anyon (continuous exchange phase $\eta=e^{i\theta}$) in a GENUINE 3+1D system — not an effective-2D FQH surface — refutes it. Matter-sector prohibition; general QFT permits anyons only in 2+1D	$\pi_1(\text{config space})=S_n$ in $d=3 \Rightarrow$ only 1-dim reps $\eta=\pm 1 \Rightarrow$ anyons prohibited; anyons require $d=2$ (braid group B_n)	effective-2D FQH anyon phases empirical, not load-bearing
	$L_2=0.000000$ (5/5, exact byte-identity) $\Rightarrow$ $M_1=0 \Rightarrow$ capacity=0; ED locality confirmed as exact channel-zero	locality confirmed empirically, not derived from primitives
	slipping requires uncommitting a polarity (P11 forbids); $\pi_1(U(1))=\mathbb{Z}$	local Coulomb withheld at the discrete layer (Grounded/Open)
	Both factors ED-structural: $1/2$ from $\kappa=1/(2r_s)$ (b-profile slope, GR-III); 2π from near-horizon Rindler shape via smoothness (numerically confirmed)	tool = Euclidean continuation, the SAME route GR uses (on par with GR, not beyond); b-profile + κ inherited from GR-III; arrow-native continuation-free 2π OPEN (§4b)
	parity-cleanness forces $B=SAS^{-1} \Rightarrow$ even det $\Rightarrow$ zero winding (numerically $N=1..6$)	transport-level ONLY; the relativistic substrate->Dirac descent is OPEN (§8); the casting is inherited
	dimensional analysis; the $I_{P^{d-1}} \leftrightarrow$ area identification uses Paper_027 §3	
	arrow (P11/P13) in the law $\Rightarrow$ preferred foliation $\Rightarrow$ mode count 2+1	D-structural (not a completed linearized propagator)

Falsifier / Kill-Cond	Derived	Inherited/Open
	forced by P05 transport typing + P04 extensivity; exact for dust	pressure/anisotropic-stress need full rule
	one substrate, one transport (P05), one cone; shared by all species	F-independent
	reserve sector dissipative (P11), not kinetic => no second cone	leading order; sub-leading reserve renormalization benign
	khronometric PPN numerator vanishes on the scalar-luminal surface once $c_T=c$, $c_s=c$ — the SAME causal cone as GW170817's $c_T=c$ (one fact, two faces)	literature-verified (Blas-Lim, Ramos-Barausse, Foster-Jacobson) + numerically checked
	metric stiffness always-on (P02); khronon stiffness only where commitments fire (P11); dense commit = Zeno => no QM, and ED has a QM sector	prefactor $O(1)$ tied to $\kappa/D=8\pi G$
	ball-cut exponent = $d-1$ (holographic surface count); 3D uniquely reproduces GR-I	derived conditional on the surface-count; metric is KINEMATIC/static-linear, nonlinear OPEN; exactly-3 is internal coherence, NOT a proof ED forces 3D
	non-abelian $SU(N)$ forced by N indistinguishable channels + bandwidth conservation (P04) + invertibility unitarity	SM correspondence $\{1,2,3\}$ and couplings inherited
	committed record = recorded structure not kernel content; channel-free (A1) + no-erasure (P11)	absolute $W_{\max}$ inherited
	$\langle p \rangle \rightarrow 0$ for a massive scalar with $\theta\text{-dot} \sim m_R$	
	cos kernel analytic => gradient expansion gives only analytic powers => canonical $p \sim 1.99$ on the real V5 functional; no $(\text{grad } \theta)^{3/2}$, no a_0	derived + numerically confirmed; one open edge = non-perturbative/vortex sector (minor)
F5/F2: the relic's conserved commitment number shown NOT ED-native -> the a^3 headline collapses, relic loses its edge over a modified-gravity dial	Noether current of the global $U(1)$; $\text{div } J = 0$ IS P11 (commitment-irreversibility) => $n \cdot a^3 = \text{const}$	derived GIVEN the field content of §2; the volume memory a modified-gravity dial structurally lacks
	carrier form from P01-P04 + P09 (audit = D); $ P_K ^2 = b_K$ algebraic	sqrt-convention + $\arg = \pi_K = P$; inner product = I (Paper #3)
	frequency limit $f_K = b_K / \sum b \rightarrow \text{Born}$ (audit D)	rests on P-LinRate (P, load-bearing) + frequentist interp (A->assumption)

Falsifier / Kill-Cond	Derived	Inherited/Open
	form FORCED by P04+P05+P09+interference (3 args)	b_K, p_{iK} values inherited
	form M3-FORCED from P04+P09+composition	Tsirelson bound $2\sqrt{2}$ INHERITED (Landau-Cirelson 1980); uses $P_K = b_K e^{i p_i}$ ($ P ^2 = b^2$), inconsistent with 001/003.5 corrected $\sqrt{b}$ (Staleness #4)
	form-consistent under DCGT + P-ScaleSeparation	measure conventions inherited
	post-state/completeness follow; Born prob D-conditional on P-LinRate	P-Commit-Projector-Match (P) load-bearing; rests on P-Channel-Orthogonality (from 004)
	M3 form-IDENTIFIED (zero pure-D rows)	fringe geom/ $\lambda = h/p/\text{visibility}$ inherited; Θ_{ED} value + V5-injection-rate OPEN; Θ_{ED} 'P12 ED-threshold' mislabel corrected 2026-07-06
	norm-preserving $\Rightarrow$ unitary; Stone $\Rightarrow$ Hamiltonian form	P-Norm-Preservation (P) load-bearing; Stone+polarization = I; specific H inherited
	load-bearing = the substrate identification (P11 $\rightarrow$ norm-preservation, P13 $\rightarrow$ time-translation)	Schrodinger eqn itself = P (definitional, Paper_095 §2.3); Stone = I; $\hbar$ inherited
	factor-2 form-FORCED via the NR limit of Klein-Gordon	independent substrate-Galilean-Jacobian derivation of $1/2 = \text{OPEN}$; $V(x)$, m inherited
	leading-order unitary form forced in-regime	V1 $\rightarrow$ unitary coarse-graining OPEN (asserted via DCGT analogy)
	emergence D-conditional on P-Motif-Algebra (P) + Cauchy (I)	inner product = I; the I $\rightarrow$ D relabel needs both Gleason postulates (Staleness #1)
	form-FORCED by P09+P10 + 3 postulates	2π normalization inherited; O-Phase-1..4 open
	polar-decomp invariance lemma + identification with U(1) gauge	substrate derivation of U(1) gauge invariance from P09 alone OPEN; role of P11 OPEN
	form-identified via Stone on the P03 translation group; CCR $[x, p] = i \hbar$ follows composition D-via-I	P-MomentumIdentification declared (substrate derivation OPEN); $\hbar$ INHERITED
		bound INHERITED (Robertson 1929 + Cauchy-Schwarz + CCR); $\hbar/2$ inherited; P-FourBand declared, substrate-derivation OPEN
	D-via-I: KL machinery applied to P-V1-Spectral + P11 retardation	$\rho(m^2)$ values I; multi-point functions OPEN
	D-via-I: QFCS + Hadamard machinery applied to V1	curvature corrections I; backreaction / Einstein eq OPEN

Falsifier / Kill-Cond	Derived	Inherited/Open
	DCGT coarse-graining + P-Polarity-Connection-Match	groups, couplings, local-gauge invariance all I
	from V1 Lorentz covariance (Paper_089) preserved by DCGT	
	P-CONSTRUCTION matching the standard form (round-8 relabel: was D, now P) under P-Scalar-Match + P-Lorentz-Covariant-Continuum	mass values I; canonical quantization / interacting QFT NOT claimed
	P-constructions matching standard forms under P-StructureConstants-Match	groups I; mass gap / OS NOT claimed here
	standard variational step FROM the postulated action	action = P (P-YM-Action-Coarse-Graining); g_YM value I
	form FORCED given postulates; new content = substrate adaptation of Perron-Frobenius/Glimm-Jaffe to V1/V5	coeff INHERITED; continuum survival conditional on P-Profile-Rescaling (OPEN); numerical Delta OPEN
	channel-count on a closed 2-surface	P-Codim-1 + P-Sat postulates; I_ED value inherited
	holographic source-resolution; N(R) factors cancel => $\Phi = -GM/R$	P-Potential-Reading (Model A); V1 1/R envelope inherited (DCGT)
	enumerates WF-1..6 with falsifiers	WF-3 conditional on the P14 placeholder; sufficiency for curvature = A->assertion
	eikonal/stationary-phase => $S = -mc \cdot \text{integral } d\tau$ => geodesic eq	FORCED-conditional on P-FreeChain + guardrails (035); eikonal->action step OPEN; M3
	opens Arc ED-10	three P-postulates; 'conditional-positive'; deep-MOND superluminality A->cost (re-tiered to 'feature' by KM line, Staleness #4)

Falsifier / Kill-Cond	Derived	Inherited/Open
	$\mu(x) \rightarrow x$ + field eq + spherical geometry; AQUAL Lagrangian variational	FORM-FORCED given P-MOND-Interpolation / P-MOND-Field-Form; μ form inherited empirically
	c invariant x H0 invariant	P-H0-Cosmological-Invariant postulate
	form-IDENTIFIED (V1 finite 2nd moment + curvature coupling + R_H domain); live-horizon $\Lambda \Rightarrow w=0$ /dead, so only the frozen floor accelerates	value inherited; M3 conditional on Route-A; RESIDUAL OPEN: force vs stipulate the frozen value (§3.5, updated 2026-07-16)
	min-structure follows once the composite is postulated	rests on P-Mcap-Unification + P-Mcap-Finiteness (postulates)
	categorization by primary driver	formal mathematical INDEPENDENCE NOT proven (own audit: A->assertion)
	3 constituents -> 3 classes; 'exhaustiveness is FORCED'	rests ENTIRELY on P-Three-Constituent + P-Always-Binding (postulates), conditional on the 13-primitive ontology + V1 + V5
	exponential form conditional on P-Rigidity-Gap + P-Boltz-Analogy	both postulates BY ANALOGY to standard topological-QM, NOT derived; leading-order form NOT unique to ED (topological-QC predicts the same)
	plateau from the finite correlation budget	conditional on P-Corr-Budget + P-Redundancy-Mapping (linear = A->regime; domain settled by 058b)

Falsifier / Kill-Cond	Derived	Inherited/Open
	structural: 058's P-Corr-Budget is per-locus, V5 pairwise/finite-reach, a topological web is local (bounded degree, no long-range pairwise load); syndrome records P11-committed (budget-free)	conditional on three inherited/QI rows
	composition structure	rests on P-Composition-No-New-Class + P-Substrate-Shift-Locality; the four mappings individually postulated
	same-threshold identification	rests on P-Mcrit-Unified + P-Cross-Platform-Universality; values inherited; phase-transition comparison is A->analogy
	D conditional on V5 (Paper_090)	
	D (standard SVD applied)	conditional on P-V5-Schmidt-Generic
	form D (form-forced) from V5 finite budget + P04 additivity	conditional on P-V5-Budget + P-Measure-Saturation; W_max inherited
	Lock L1 (marginal independence) = D-via-I from tensor-product + POVM completeness	L2 rests on P-Substrate-Causality; L3 = Paper_073; '3-locks => over-determination' is A->position
	D-via-I (Shannon-Khinchin + Khinchin uniqueness)	conditional on P-Continuity + P-Additivity + P-Maximality; normalization inherited
	from P02+P04+P07+P08 + UV-finiteness	leans on an explicit P-SubstrateLocality postulate
	structural exclusion via P11	rests on P-NoBackwardChain + a substrate-operational (NOT algebraic) closure; M2; upgrade to M1 needs P11 from a deeper layer (OPEN)
	D-via-I	inheritance postulated (P-Causality-Propagation); specific arrow derivations OPEN
	dichotomy FORM (only eta=±1, fermion/boson exhaustive) from P06+P09 exchange topology + $\pi_1(SO(3))=Z_2$	specific spin assignments (e=1/2, gamma=1) empirical; CPT/parity/T deferred
	structural mechanism identified: unique first-order Clifford factorization of KG on the Cl(3,1) bundle; §3.7 continuum limit COMPUTED + Wilson-term undoubling 16->1	explicit substrate-V1->Dirac closed proof OPEN; fully-covariant proof OPEN; graph-native SPINOR construction OPEN (canonical P07 supplies NO native channel topology); mass inherited
	substrate IDENTIFICATION of m as P04 rest-bandwidth; V1-kernel low-momentum-pole compatibility	KG form ITSELF is P-definitional/INHERITED from Paper_017 + std dispersion; V1->KG-propagator OPEN

Falsifier / Kill-Cond	Derived	Inherited/Open
	extension of KG to U(1) gauge; Noether current from U(1) phase-invariance	minimal-coupling form INHERITED (015/016); couplings empirical; V5 multi-channel substrate-graph derivation OPEN
	finite SL(2,C) (j1,j2) taxonomy as substrate rule-type classification	which rule-type $\leftrightarrow$ which rep empirical; Wigner particle-state classification out of scope
	UV-finiteness from P04 finite bandwidth + P08 scale + V1/V5 finite-kernel form factors	cutoff $\omega_c = c/l_P$ value inherited; substrate-graph $\rightarrow$ power-counting derivation OPEN; renorm-reframe = A $\rightarrow$ position
	null-geodesic propagation form-forced from massless dispersion (T5 at $m=0$) + std null-geodesic geometry	EXISTENCE of $\sigma=0$ rule-types CONDITIONAL on Paper_114; substrate-c inherited; V1 at $m=0$ derivation OPEN
	3 structural features form-forced: Lorentz-scalar, P04-bandwidth-anchored, statistics-class-mechanism	substrate-Higgs §3.3 + substrate-Yukawa §3.5 ASSERTED, NOT derived — OPEN; all mass values empirical; origin of 'statistics-class-mechanism' OPEN
	existence form-forced CONDITIONAL on P-Gauge + T17 (015) + gauge-quotient (022)	specific gauge groups/couplings inherited; P-Gauge substrate derivation OPEN
	vacuum + 1-particle Fock decomposition from T10 + T8 + Wightman uniqueness	'no rest-frame $\Rightarrow$ minimal dual' is heuristic, OPEN; mode-density normalization inherited
	two-branch mutually-exclusive massless slot (fermion-chiral MR-R / gauge MR-P); disjointness forced by T7	no-third-mechanism exhaustiveness OPEN; occupants empirical (neutrino MR-R); substrate-c inherited
	bandwidth-collapse rule + 'weakest-link' (minimum) endpoint seals every boundary edge	the bandwidth-collapse rule's admissibility inherited from the gravity arc; one construction
	discrete holonomy $\sum d\phi = 2\pi w$; standard int $(\text{grad } \phi)^2$ electrostatic action	local Coulomb field withheld at the discrete layer (OPEN); DCGT-selection = the honest open edge
	3 standing pillars: (1) I_ED-fine dynamics = coherent Schrodinger evolution (emergent unitarity), not measurement; (2) einselection environment-set + coarse (energy eigenbasis for bound charges); (3) commitment individuates only distinguishable channels, distinguishability operational (A1)	explicit ED theorem + exact distinguishability floor OPEN (downgraded from load-bearing to write-up)

Falsifier / Kill-Cond	Derived	Inherited/Open
	standard $\gamma^5 = i \gamma_0 \gamma_1 \gamma_2 \gamma_3$ read in ED terms (both factors native); vector theory carries no gauge anomaly	chiral cancellation inherited with the content; contingent on the substrate $\rightarrow$ Dirac descent (open)
	FORCED jump-operator identification + exact projector algebra (a structural result); resolves the P11 line-gap 'C3 tension' (committed = dark states)	NOT a distinctive observable — standard QM predicts the same null (immunity, not a win)
	from einselection (definiteness) + the finite substrate grain (regulator)	
	the scaling + its deeper origin (the arrow supplies Penrose's missing time)	E_G is IDENTIFIED with Penrose's self-energy, NOT computed (honest flag C4); coefficient OPEN; exact energy functional OPEN
	P-RB-1 + P08 + Paper_027 + Paper_029; masses inherited	
	corpus (Paper_027) + dimensional analysis	
	ties Paper_027 §3 to Paper_MetricFromTheGraph (bandwidth=boundary cut $\sim R^{d-1}$)	
	exp growth via Q1A+Q2A+C3 chain (row 13 CLOSED)	Friedmann/de Sitter step I; uniform-Psi IC-inherited; Route A open
	catastrophe-dissolution D-via-I via Paper_038_5 §3.7; $\rho_L = (3/8\pi)H_0^2 M_P^2$	smallness = quantitative-magnitude conditional on Route A (single load-bearing OPEN)
	$w = -1$ via the SAME inherited Q1A+Q2A+C3 chain as Cos-01	Lambda value + $\Omega_L \sim 0.685$ I; H_0 pending Route A

Falsifier / Kill-Cond	Derived	Inherited/Open
F3: MOND scale shown CONSTANT with z refutes it. First data (MUSE-DARK III, A&A 2026) instead EXCLUDE constant a_0 at $\sim 30\sigma$ (evolution confirmed); a robustly-confirmed rate incompatible with $\sim H(z)$ refutes the EXACT bet (data run faster), not the evolution DISCRIMINATOR (2026-08-02): $a_0=cH(z)$ [ED, RISES with z] vs $a_0=c_2\sqrt{\Lambda}$ [CONSTANT] coincide today but diverge with z; a_0 following the $\sqrt{\Lambda}$ (Λ)-constant form rather than $cH(z)$ refutes the ED relation. Recent high-z RAR literature (arXiv 2405.01841, 2504.20857) shows a MILD PREFERENCE for a_0 rising with $z \rightarrow$ favors ED's $cH(z)$ over the Λ-tied form.	unfudgeable SHAPE (horizon projection introduces no I_P; no free dimensionless number to bend the power)	evolution CONFIRMED $\sim 30\sigma$ (MUSE-DARK III, A&A 2026), excludes constant a_0 / kills constant-scale MOND. **2026-09-07: the $\sim 30\sigma$ does NOT separate ED from LCDM** -- Magneticum (MNRAS 10.1093/mnras/stac3017) shows plain LCDM WITH BARYONS produces an apparent rising a_0 from galaxy assembly alone, no modified gravity. It excludes constant-a_0 MOND and nothing else. EXACT rate: tension now COMPUTED, not estimated -- $\alpha = 1.15 \pm 0.07$, $\alpha=1$ at 2.3 σ (event-density/theory/a_0z_powerlaw_refit.py); the published ± 0.04 omits the intercept, cosmology and reduction choice = 73% of the variance. Survey systematics ON TOP and unquantified. NB their fit is LINEAR: $\alpha = 1.18$ was always an ED-side recast, and their line has no constant exponent (local slope 1.78 $\rightarrow$ 0.90, crossing $\alpha=1$ at $z \sim 1.1$). 2026-09-07 ROUND 3: the systematics challenge is REAL and located - Rusishvili 2026 (academia.edu/170831186, self-posted, NOT peer-reviewed) re-analyses Ciocan's own public catalogue and reports three baryonic systematics, each claimed sufficient. Effect sizes are QUANTITATIVELY COHERENT (checked: theory/a_0z_baryonic_systematics_check.py - the claimed 82%% from molecular gas falls out at $M_{HI}/M^* \sim 0.1$, the expected regime, with no free parameter). It attacks the $\sim 30\sigma$ DETECTION, not just the rate. If it holds, the qualitative win AND the 2.3σ tension go together and the honest status is UNTESTED. DO NOT BANK EITHER.

Falsifier / Kill-Cond	Derived	Inherited/Open
		form-complete / native — neutral-channel sibling of the baryons (Baryogenesis) + binding mass (Mass-Without-Mass) + channel/polarity (MS-II)
	O(1) hierarchy $g_{int} < g_{ext} < a_0$; EFE-class suppression from finite outer-scale capacity	$a_0 = cH_0/(2\pi) I$ (Paper_029); the suppression FUNCTION not yet derived from primitives (§5)
	regime partition form via SC-4.9 Hessian + Morse theory + V1/V5 composition	I_{saddle} substrate-parameter-INHERITED (Route A4); SC-4.9 exhaustiveness OPEN upstream
	LDE/saturation case substrate-graph-IDENTIFIED directly; RDE/MDE form proposed	RDE/MDE form inherited (QFT analog); Friedmann/ H_0 inherited; Q1+Q2 load-bearing OPEN
	Larmor + quadrupole FORM via the NoetherFlux chain	numerical coefficients INHERITED; OPEN-RL-Q1 (source-class) + OPEN-RL-Q2 (flux form) load-bearing
	collapse mechanism form via Dyn-01 S1 + Dyn-02 + GW-01	monotonicity itself inherited from NR phenomenology (substrate-graph proof NOT constructed); $r_{horizon}$, Choptuik γ inherited
	two-saddle form via Dyn-01 S2 + Dyn-03 + Dyn-04 + GW-01	Peters 1964, PN coefficients, r_{ISCO} inherited; plunge = NR domain
	given C-amplitude (from the keystone) + total bandwidth (P05 isometry); tree-level lift verified (0 for U(1), lifted U(2/3/4))	indistinguishability alone gives only S_N ; continuum survival = the Clay core, OPEN, NOT a Clay proof; {1,2,3}/anomalies/charge-norm NOT derived
	operational distinguishability ($\max \langle K E K \rangle = 1 - c^2$); commits only in the channel basis; P09 rules out R, composites rule out H	the Soler lattice technical rigor is the residual (OPEN, #8b Gate 1)
	front-null + commitment-rate law $\Rightarrow N^2 \sim b$	D-cond on P-Commitment-Linear ($\alpha=1$) — later FORCED in GR-III §4
	$n_{opt} \sim b^{-1} = \text{square of spatial index (sim 2.09 vs 2.00)}$	conditional on 13 primitives; sim uses imposed Newtonian-limit b-profile
	substrate mechanism; the 2 tensor modes = the 2 independent Hessian-flip directions (§3.7)	speed=c, strain, frequency, 2-pol count all INHERITED; full waveform (row 13) OPEN non-load-bearing
	substrate Noether flux $\rightarrow$ DCGT $\rightarrow$ linearized-Einstein form	coefficient $1/(5c^5)$ INHERITED; row-12 'closed' = FORM-only, coefficient OPEN (Staleness #2)
	ringdown mechanism + discrete-mode-origin form via Dyn-01 + GW-00 + Dyn-02	QNM frequencies $\sim c^3/GM$ INHERITED (Schwarzschild/Kerr; Regge-Wheeler/Teukolsky); IC from NR merger

Falsifier / Kill-Cond	Derived	Inherited/Open
	integration form via GW-00+GW-01+Dyn-04+Dyn-05+Cos-01+Cos-05	population synthesis, H(z), H0 inherited
	metric-borne modification; unitary-gauge no time-derivatives	slip O(Phi)-suppressed
	promoting the acceleration term to $W(A^2)$ => AQUAL in the metric potential; deep-IR embeds 030	single hinge = 'forced-given-030' (matches the derived Combination Rule, does not re-derive); interpolation a Cassini-constrained family
	kinematics of static congruences	the cleanest result; sequesters the cosmology sector
	mapping	class mechanisms are DIRECTIONS, not verifications; abundance = value-inherited IC
	shared SCBU boundary; scaling exact; sign passes (V1 density positive => $\Lambda > 0$)	D-via-I (tier transported from 038.5, not upgraded); ab-initio integral OPEN with a numeric target
	handed-commitment twist as the chirality locus	evidence is a 1+1D TOY-lattice computation, NOT canonical V1; γ^5 relocated to GLOBAL (=i g0 g1 g2 g3); 3 open bridges to a relativistic result
a stable/observed gauge sector requiring $SU(N \geq 4)$ — a 4th independently-stable channel family in the internal amplitude space — refutes the d=3 reduction	max mutually-orthogonal channels in C^d is $N=d$	WHY d=3 is itself OPEN
	orientation-entanglement / Dirac-belt reading	$Cl(3,1) + \pi_1(SO(3)) = Z_2$ inherited (T2/T4)
	P05-transport = U(N) link variable; relational graph (no Brillouin torus) + retarded arrow (non-hermitian) evade the two N-N premises	'P05 re-routes channels unitarily' is a structural reading, not a closed proof
	topological elimination (a link holds in 3D, frozen in 2D, unravels in 4D)	conditional on the OPEN premise that ED holds order by spatial linking

Falsifier / Kill-Cond	Derived	Inherited/Open
general gauge theory PERMITS chiral abelian U(1)s; ED forbids them. A measured parity-violating abelian (U(1)) force refutes it. Matches the low-energy world (EM abelian=vector, weak non-abelian=chiral)	V1 single-channel chirality-blind -> abelian vector; V5 cross-channel chirality-sensitive -> non-abelian can be chiral	WHICH force is chiral (weak assignment) OPEN/inherited
	the EFE derivation (structural) + the Newtonization arithmetic (solid); the naive 5-50x is an EFE artifact	conditional on P-Quadratic-Strain (Model C postulate) + the constructive interference sign (MEASURED tier only)
	forward arrow = P11; controlled inter-region capacity = EXACTLY zero (A1, measured)	
		structural (standard condensate/Madelung form + corpus field content)
		structural — the two-speed split is the arrow's preferred frame, not chosen by hand
	the two mechanisms (D-via-I rows 7-8); joint H0-scaling (D rows 9-10)	numerical a0/xi values I (row 11); xi_canonical-from-H0 OPEN (the paper's single open item)
	Paper_090's difference-phase gauge law forces the conjugated-pair moment; Coh maximal at alignment	conditional on P-Bipartite-Mapping + like-chain reading; the earlier TSIRELSON characterization was RETRACTED (line 93), replaced by this gauge-law route; reach inherited
	same int K_V5 at the P08 grain (structural identity); P-V5-Budget + P04 additivity	absolute W_max inherited; path-topology not capped per-locus
	D-via-I	faithful substrate-graph confirmation 'precluded on existing tooling' (in-queue)

Falsifier / Kill-Cond	Derived	Inherited/Open
		M3 form-IDENTIFIED + value-INHERITED; delta T/T, eta^G, 4piG all I; mix of A->position + D-via-I rows. 'Does not commit ED to cyclic cosmology'
		A->position, conditional on the contrast ontology (Facts paper) — grounds CCC's assumption, not a substrate derivation
	H(t) via continuum-Friedmann (D-via-I); freeze-out sequence D-via-I	Y_p,D/H,3He,7Li, nuclear physics, eta_b I from standard BBN codes
	r_s,D_A,peaks,baryon-loading,Silk D-via-I via continuum-Friedmann	peak positions/heights/TE/EE I from CAMB/CLASS; n_s, Omega_b h^2 IC-inherited
	D(a),T(k),k_eq,BAO r_s D-via-I	sigma_8,P(k),Omega_m h^2 I; §3.5: 'ED supplies no late-LDE suppression BEYOND LCDM'
	P_zeta,P_T,n_s-1=-6eps+2eta,r=16eps D-via-I via standard slow-roll	A_s~2.1e-9,n_s~0.965,r<0.06 I from Planck+BICEP
resistivity keeps rising with no saturation once loss controlled (MIR already observed -> consilience pass, not a new bet)	form-forced given P-Scattering-Budget-Saturation	ceiling value rho_max INHERITED (as phi_max, n for viscosity)
	10 seeds x 40 snapshots (400 total), 64^2 periodic, via the simulator of record r2_grf_falsifier_tests.py; GR-SC 1.7 half-decay of the 2D radial autocorrelation	40 seeds x 40 snapshots; per-snapshot spread 0.334, per-seed 0.179, standard error 0.028; canon-INTERNAL (not derived from primitives)
	build of 40k loci; short-range->r^2.02, fully-long-range->r^2.7 (could-say-no fired)	assumes the emergent geometry (does not derive it); reproduces a known result = consilience, not a novel prediction
	fixed point of the two forced terms; horizon forms with nothing about a horizon in the rule	1/4 coefficient inherited
	vacuum solution + GR-I g00·grr=-1; corrects an earlier 'flat' measurement error	direct sim resolution-limited
	measured convergence	2 of 3 counts assume the emergent geometry (curvature emergence OPEN); scale I_V5~I_P inherited
	diffusion R^2~0.001; ballistic v ~1; pure ED never mixes (CV~1.2 vs 0.00); front mobility concentration-independent	certified Sigma-substrate only; UDM is a separate posit (refutes UDM-from-these-dynamics, not UDM)
	blind->common-cause only; no ordering coupling anywhere (5-arc); within-cluster spread 0.084->0.0005 at reach<D	structural reading, NOT a theorem; silent on topological defects (knots); V5+memory honest structural additions

Falsifier / Kill-Cond	Derived	Inherited/Open
	coupling-off disperses (28->55), on confines (1.4-2.3); forced-response 0.72 vs unbound 0.97	V5-CONDITIONAL (a structural addition, not primitive-forced); binding mass only, fundamental Higgs INHERITED
	lattice $\int(\text{grad } \phi)^2$: $p=1$, $R^2=0.97$ (vs 0.77 at $p=2$); substrate kinetic + phase-sector Sigma-blind	analytic/computed, NOT a substrate measurement — confirms the identification only
	1+1D toy: point-gap winding, skin effect, spectral flow — all zero in the Hermitian control	not yet Lorentz-covariantized to a γ^5 coupling
	linking_3d_vs_4d_probe: unlinking forces a collision (0.000) in 3D, slides free (0.600) in 4D	the ED-holds-order-by-linking premise still OPEN
	trivial connection crystallizes ($R=1$); disorder+irreversibility $\Rightarrow$ xi a few lattice units; within-cluster spread $\sim 0.27x$ at finite reach - all on Grad intensively scored	reach xi value-inherited AND functional-dependent; V5 existence stays a P10 posit; canonical Coh shows NO alignment gain under any normalization tried (#111), so the finite-reach result does not transfer to it
	build-verified 2D/3D on Grad intensively scored; not crystalline on ALL THREE functionals (ten seeds, 0/10)	the SIGN transfers from the P12-Coh paper; the build behind it measured Grad, not Coh
	V5 finite-reach retarded attraction confines fronts into a sub-luminal bound composite	mechanism native but V5-CONDITIONAL (V5 = faithful structural addition, not shown primitive-forced); mass VALUE inherited
		VERIFIED NOT BANKABLE (2026-07-26): the winding->phase-circulation map is UNMADE — ED's B4 winding is the CHARGE phase (P09 on graph cycles), NOT the Madelung wavefunction phase Wallstrom is about; 3-step identification burden all absent; also gated on the Gleason #8b open step
		OPEN — 023 names all three and states the arc 'does not close' them
		OPEN

Falsifier / Kill-Cond	Derived	Inherited/Open
		OPEN (§3.5) — bootstrap knot; the driver is matter-dilution (Cos_05)
		OPEN — highest-leverage open derivation in the program
		OPEN
		OPEN
		OPEN
		OPEN — target #4; 051 assumes it (P-Substrate-Unitarity)
22-Ways P2 (merger-lag): PROVISIONAL, existence-only; NOT refutation-grade (needs substrate tau_V5 floor, OPEN)		OPEN
		OPEN — A/B/C distinctness + exhaustiveness inherit only 'distinct primary drivers'
		OPEN (A->regime) — alternatives (exponential, log) are substrate-consistent and would change M_cap AND the wall; load-bearing under both the wall's shape and number
		OPEN — acknowledged retroactive-fit risk until closed; only the sharp/architecture-independent FORM is retroactive-fit-immune today
		OPEN ('in-queue'; currently by analogy to standard topological-QM)
		OPEN — the live experiment; NO confirmation claimed
		OPEN
		OPEN
		OPEN — structurally blocked; ED has no substrate- level temperature
		OPEN
		OPEN
		OPEN

Falsifier / Kill-Cond	Derived	Inherited/Open
		resolved NEGATIVELY for the determinate substrate (kinetic + Sigma-blind), so the smooth field is a shadow; the coarse-grained-selection question named
		OPEN — derived-2pi route is Euclidean (reversible-time), on par with GR; arrow-native 2pi from raw commitment statistics not yet supplied (§4b honest negative; possibly a category error — 2pi may be an intrinsically continuum feature)
		OPEN (three named; rows 14,15,17)
		OPEN (a write-up task, no longer load-bearing)
		behind a fully relativistic chirality statement + the one non-inherited anomaly candidate
		OPEN
		OPEN — the quadratic-strain interference cross-term could shift it
		OPEN — the single load-bearing OPEN in each; 'highest-leverage open derivation in the program'
		OPEN, non-load-bearing — inherited standard-cosmology puzzles
		OPEN -> #5b. Mimetic-DM ruled out 3 ways; UDM-at-recombination ruled out (crowding $c/c_{\text{max}} \sim 1e-114$); horizon-boost baryon-coupled (wrong type); only working option = added AeST-class field. 'Native and open, not a bolt-on'
F6: a computed CMB spectrum from the relic action that misses the Planck peaks for EVERY mass in the warm window		(see Falsifier F6) NOT DONE — the largest owed piece; well-specified AeST-style Boltzmann run, now gated only on the mass/abundance
		OPEN / unaddressed — ED has engaged neither (horizon jurisdiction, no mechanism worked out)
		OPEN — not derived, only order-of-magnitude estimated
		OPEN — order-of-magnitude only; exact knee shape uncalculated

Falsifier / Kill-Cond	Derived	Inherited/Open
		load-bearing (Q1+Q2 joint closure -> M3)
		load-bearing; closing Q1+Q2 lifts Dyn-02/03 (and by propagation Dyn-05, GW-02) M2->M3 in one move
		the load-bearing residual riding under the whole reconstruction (#8b Gate 1)
	channel-orthogonality REDUCED to channel-distinctness GIVEN the representation (operational distinguishability); C selected; covering law candidate-grounded; einselection primitive; Born non-contextuality resolved; equal-norm/angle discharged 2026-07-10	residual = Soler lattice technical rigor ONLY
		OPEN — the bridge under 2-of-3 tiling counts + the Kerr geometry
		OPEN — now carried by a separate two-component warm relic (One-Field + 117 banners); see ../dark-sector/
		OPEN — the nonlinear regime is the distinctive/falsifiable frontier (kinematic metric only so far)
		NOT derived — explicitly deferred to OPEN (archived-source check CLEAN: no four-band/dwell-mass reliance)
		OPEN — reduces to the T4 spinor gate #2b
		OPEN — reduces to the T4 spinor gate #2b (highest-leverage open item in the program)
		OPEN (#5c) — substrate deposit field measured non-Gaussian below ~32 cells; reproducing known CMB objects does not pay it. DISTINCT from the #5b dark-matter debt
		OPEN — the live risk; mechanism gives cold GeV-Planck, needed warm ~keV (6-26 OOM gap). If forced cold, relic clumps in galaxies and the sector fails

Falsifier / Kill-Cond	Derived	Inherited/Open
		OPEN (minor) — the gradient expansion covers only the perturbative kinetic sector; a defect contribution is the sole residue of the one-substance route
		OPEN — the paper's single open item; would upgrade M3->M2
		V5 existence stays a P10 posit
		P-QMkin-Composition — 'not from P02+P03 alone... a paper-specific postulate' (preamble #2)
		'a commitment, not a set of derivations'; hbar (O-RB-1) and c-from-deeper (O-RB-2) OPEN
		A->analogy; SUPERSEDED/upgraded to Derived by MS-II §2 (Staleness #2)
		the coarse-grained action = standard S_YM (the substantive declared content)
		the one non-inheritable OS axiom (the rest of OS inherited)
		the Clay-hard wall; underived — the load-bearing OPEN gap for continuum survival
	scale ~cH0 = D (Paper_029 audit row 12, reached three independent ways); the 2pi is NOT (audit rows 8, 10, 11)	a0 value inherited via H0; ~10% POSTDICTION; coefficient DISPUTED (gravity Staleness #10)
		postulated, non-unique (§8.4/8.9); later DISCHARGED by QuadraticStrain as the forced interference modulus
		all six are P (postulates) with substrate-consistency justifications
		P-Bipartite-Mapping — 'definitional commitment, not a separate derivation'
		P-V5-Schmidt-Generic — derivability from V5 envelope alone is OPEN
		P-V5-Budget, P-Measure-Saturation
		P-V5-Hilbert-Constraint
		postulated NOT derived; explicitly NOT claimed minimal ('necessary-for-current-derivations', not absolute)

Falsifier / Kill-Cond	Derived	Inherited/Open
		redundancy among the three OPEN
		derivability from the other 12 primitives OPEN
		derive-from-13 vs elevate-to-14th OPEN (discharged by QuadraticStrain per the gravity ledger)
		derivability from the 13 mostly OPEN; count inconsistent across papers (Staleness #3)
	budget + irreversibility + retardation -> forward-causal	declared as P-Primitive-Forward-Causality; composition D-via-I
	gauge-class connection A_μ on rule-type bundles + gauge-quotient	substrate derivation from P05+P07+P09+P10 alone OPEN (F-Gauge would upgrade M2->M1)
	P04 reserve monotone-draining (P11) => $\beta=0 \Rightarrow \alpha=1$	D-cond on a named band-accounting premise
		naming convention / definitional ONLY per Paper_095 §2.3; no new substrate content
		one postulate replacing P14 + 030's regime assumption
		argued CONSISTENT-WITH, not forced by, Paper_058's Class-C accounting; falls with F-QD-1
		P-Substrate-Cosmology-Unified (the single new postulate; audit row 6)
		P10 licenses without forcing; forward derivation open
slope != 4 or irreducible intrinsic scatter (data 3.95+-0.08, ~0.1 dex). MOND-SHARED: discriminates ED+MOND from LCDM; only the a_0 normalization is ED-distinctive		(see Falsifier)

Falsifier / Kill-Cond	Derived	Inherited/Open
departs from $cH(z)/2\pi$. Only 031 §8.8 is data-confronted: MUSE-DARK III (A&A 2026) $\sim 30\sigma$ evolution, $\sim 8\%$ local, edge of $H(z)$; $\alpha=1$ the decisive open number. Do NOT over-bank (direction won, exact rate tense) DISCRIMINATOR (2026-08-02): $a_0=cH(z)$ [ED, RISES with z] vs $a_0=c_2\sqrt{\Lambda}$ [CONSTANT] coincide today but diverge with z; a_0 following the $\sqrt{\Lambda}$-constant form rather than $cH(z)$ refutes the ED relation. Recent high-z RAR literature (arXiv 2405.01841, 2504.20857) shows a MILD PREFERENCE for a_0 rising with $z \rightarrow$ favors ED's $cH(z)$ over the Λ-tied form.		(see Falsifier); ONLY 031 §8.8 reflects the 2026 data 2026-09-07: exact rate now COMPUTED at 2.3σ ($\alpha = 1.15 \pm 0.07$), and the $\sim 30\sigma$ excludes constant-a_0 MOND only -- LCDM with baryons (Magneticum) also gives apparent evolution. 2026-09-07 ROUND 3: the systematics challenge is REAL and located - Rusishvili 2026 (academia.edu/170831186, self-posted, NOT peer-reviewed) re-analyses Ciocan's own public catalogue and reports three baryonic systematics, each claimed sufficient. Effect sizes are QUANTITATIVELY COHERENT (checked: theory/a0z_baryonic_systematics_check.py - the claimed 82% from molecular gas falls out at $M_{HI}/M^* \sim 0.1$, the expected regime, with no free parameter). It attacks the $\sim 30\sigma$ DETECTION, not just the rate. If it holds, the qualitative win AND the 2.3σ tension go together and the honest status is UNTESTED. DO NOT BANK EITHER.
activity-INDEPENDENT lensing refutes. PROVISIONAL — α_{act}, $A(x)$, the $\nabla \mu ^2$-vs-feedback discriminator observable all OPEN		(see Falsifier)
22-Ways P2 (merger-lag): PROVISIONAL, existence-only; NOT refutation-grade (needs substrate τ_{V5} floor, OPEN)		PROVISIONAL / M2, existence-only; τ_{V5} floor OPEN; numerics inherited; zero D rows (row 14 A->position); NOT refutation-grade in BTFR sense; confirms 22-Ways P2
no sharp architecture-independent wall (smooth environment/architecture-DEPENDENT decoherence) refutes it. Standard decoherence-only + CSL predict no such wall. NUMBER 140-250 kDa is PRESSURED (≥ 170 kDa interference already shown, Nature 2026, still QM-consistent) + the second-harmonic leg CONFOUNDED (KDTL optics); the SHAPE is the claim -- do NOT over-bank the number		(see Falsifier); diverges from decoherence-only (no sharp architecture-independent wall) and from CSL (tuned)

Falsifier / Kill-Cond	Derived	Inherited/Open
		the leading exponential form is ALSO predicted by standard topological-QC; the wedge is only at higher-order cross-platform consistency
surface-code clean suppression (Willow Lambda~2) is consistent-BY-CONSTRUCTION, NOT a passed test. Live pair = repetition-code floor persistence (Google ~1e-10 above d~15) + cat-state width ceiling; a floor surviving burst-mitigation at cross-platform-consistent content = the ED signature. Standard FT predicts error->0		(see Falsifier); topological branch predicts ABSENCE of plateau; surface-code suppression consistent-BY-CONSTRUCTION not a passed test; repetition-floor + cat-width pair = the live test
F2/F4: antihydrogen falls at a different rate (ALPHA-g 2023 consistent); antimatter domains / annihilation signatures		(see Falsifier)
a demonstrated LAW-LEVEL (non-spontaneous, non-contingent, fixed) origin of the weak handedness refutes it. Distinctive vs the SM's law-level fixed handedness		(see Falsifier); vs the SM's law-level fixed handedness
F1/F4: clean phantom $w < -1$ over wide z , or $w\text{-dot} \neq 0$, or $\Lambda\text{-dot} \neq 0$. Distinctive $w = -1$ null; ties 22-Ways $w = -1$		(see Falsifier)
F1/F2: A_s/r inconsistent with Route-A4 H_{infl} . CHANNEL-SPECIFIC, NOT a test of slow-roll/inflation		(see Falsifier)
F1: a cold heavy WIMP detection, or perfectly-cold DM, refutes the warm relic. LIVE RISK: mechanism prefers cold GeV-Planck (6-26 OOM too heavy)		(see Falsifier); consistency BOUND on free-streaming (~0.1-1 Mpc); keV = thermal-equivalent
F2: DM substance found INSIDE galaxies (halo, CDM cusps, scattered mass) refutes the horizon-only galactic account		(see Falsifier)

Falsifier / Kill-Cond	Derived	Inherited/Open
environmental independence (LCDM) or smooth MOND-EFE roll-off refutes; suppression function not derived (Open)		(see Falsifier); suppression function not derived (Open)
		metric-only coupling has no vector field to make a slip; Paper_038_6 PROVISIONAL
Preds 1-4: environmental independence (LCDM) or smooth MOND-EFE roll-off; knee sharper than MOND (Pred-4 uniquely ED, below current precision)		(see Falsifier); suppression function uncalculated (Open)
a clean phantom $w < -1$, or a stiff $w = 1$, outside the substrate-admitted $\{-1, 0, 1/3\} + \text{composites}$ set refutes it. NEAR-distinctive, but sits BEHIND the Q1/Q2 OPEN (currently inherited, not derived); ties to Cos-05 $w = -1$		(see Falsifier); NEAR-distinctive (substrate-side admitted set) but sits BEHIND the Q1/Q2 OPEN
definitive absence of a scalar/breathing GW polarization excluding the khronon mode		(see Falsifier)
ANY nonzero α_2 kills it ($\alpha_2 = 0$ EXACT, structural); $\alpha_1 > \sim 1e-5$ kills it ($\alpha_1 < \sim 4e-93$ safe). Passed favorably		(see Falsifier)
		all survival-bets SHARED WITH GR (not distinctive); the NANOGrav $\Omega_{\sim f^{2/3}}$ is self-labeled model-degenerate postdiction
LCDM velocity-independent scatter, or MOND-EFE smooth roll-off, refutes. Only the SHAPE is ingredient-robust; the knee location rests on a flagged $\nu \sim 0.70$ (3D O(3)) assumption		(see Falsifier); only the SHAPE is ingredient-robust

Falsifier / Kill-Cond	Derived	Inherited/Open
any measured environmental-field dependence of a collapse rate = an ED fingerprint (Penrose/Diosi carry none). But $\sim 10^{-6}$ -small near Earth (needs deep-void $g_{\text{ext}} < \sim a_0/100$, inaccessible); its absence where ED predicts it would tension ED. NOTE: non-emission of collapse radiation is NOT an ED prediction (shared with standard QM)		(see Falsifier)
no commensurate ceiling, or standard fault-tolerance error $\rightarrow 0$ with no plateau, refutes it. Standard physics has neither ceiling. D-conditional on P-QD-LiveWeight		(see Falsifier); standard physics has neither ceiling
the two onset at different substrate occupancies (no correlation); or a second inter-locus transport channel exists beyond P05/Adjacency (F-BAND)	co-onset D-via-I (form) given P-Adjacency-Transport-Shared	exact $\eta_{\text{min}}:\rho_{\text{max}}$ ratio OPEN (needs Π_p , Π_q); MIR value inherited
		postulated axiom, NOT derived
		postulated axiom
		postulated axiom
		postulated axiom (the README's 'four-band partition' descriptor is archived M-series; canonical = non-neg additive scalar)
		postulated axiom
		postulated axiom; EXTRAORDINARILY load-bearing (area law, inverse-square, all gravity geometry); derivability from the other 12 OPEN
		postulated axiom
		postulated axiom; value inherited via Newton-recovery
		postulated axiom
		postulated axiom

Falsifier / Kill-Cond	Derived	Inherited/Open
		postulated axiom; the arrow = $P_{11} + P_{13} + V_1$ retardation (T18)
		postulated axiom
		postulated axiom
	two-point/Wightman/propagator presentation of the canonical V_1 (Paper_089); adds no new primitive	
		reference catalogue, not a derivation
	the first kernel rule-type; admissible-class definition	specific envelope shape (exp/Gaussian/power-law) value-layer inherited
	the second kernel rule-type; mediates inter-chain correlation	P10-LICENSED (existence a POSIT under P10, NOT primitive-forced); forward derivation of existence OPEN; τ_{V5} values inherited
	THE verdict grammar every ledger cites	the M1 EXAMPLE (§3.1) is STALE: Phase-1 QM is M2 per 098 (Staleness #1)
		Wave-3 (2026-05) extensions; do NOT create new tiers
		INHERITED (Landau-Cirelson 1980); a standing construction — NOT the separately-retracted 'Tsirelson reduction' V_5 characterization
		value INHERITED (Paper_RQM_hbar)
		INHERITED (value-layer matching) across the whole arc
		quantization INHERITED from standard Chern-class topology; ED supplies only the rule-type-bundle identification (form-IDENTIFIED, M3)
		INHERITED (empirical matching)
		form-DERIVED, value-INHERITED (M3); $I_{ED}=I_P$ is the empirical-matching choice, not a substrate derivation
		SCALE form-derived, ~10% match; the $1/(2\pi)$ COEFFICIENT postulated/disputed (2026-09-04); H_0 INHERITED; 029's reading is STATIC (tracks whichever H_0)
		value-INHERITED; smallness = $(H_0/M_P)^2$ via Friedmann, gated on Route-A

Falsifier / Kill-Cond	Derived	Inherited/Open
		I — via log-g matching ($\log g = \pi$); 'matching, not derivation' (§4). NOTE: folder also has the geometric DERIVED route (BH_Thermal2Pi); net folder status of the 1/4 = DERIVED
		I — 'NOT a new derivation... INHERITED from Regge-Wheeler 1957/Zerilli/Page'
		I — 'INHERITED from standard semi-classical GR'; genuine ED content = V5-saturation identification only
		I
		all empirically anchored / platform-inherited
		2-POINT extrapolation (Eibenberger 2013 ~10 kDa + Fein 2019 ~25 kDa). UNDER PRESSURE: interference already shown at ≥ 170 kDa (Nature 2026), inside the window, still QM-consistent. Paper_056 does NOT record this (STALE, Staleness #1) -- do NOT bank as firm
		platform design / standard QI
		I — SVD is standard math (round-7 relabel: was D-cond-SVD, now I)
		I
	D-via-I (form composition)	entirely conditional on P-V5-Hilbert-Constraint
		I — 'not derived from first principles... INHERITED' (preamble 1)
		INHERITED (empirical, via Newton-recovery Paper_027 §5.3)
		INHERITED / matched — existence & form structural, numerical values NOT predicted
	the commutator-vs-anticommutator DICHOTOMY forced by T1 (boson $\leftrightarrow$ commutator, fermion $\leftrightarrow$ anticommutator)	relation FORMS are P-definitional; hbar INHERITED; earlier 'structural origin of hbar' framing disowned as oversold

Falsifier / Kill-Cond	Derived	Inherited/Open
	dimensional structure (action=bandwidth x time) form-forced by P04+P05+P11; chain-step invariance (P03+P13) => hbar universal	value INHERITED; B_P04 has NO independent substrate definition — §3.3 CIRCULARITY disclaimer: hbar=int B_P04 ds is a relabeling, not a computation
		inherited / substrate-parameter-inherited
		INHERITED (a single substrate parameter; substrate derivation open)
		IC-INHERITED, framework-neutral; native Z2 relocated from postulate to IC, not derived
		INHERITED (account, surviving candidate) — the SM non-monotone pattern matches no transport winding
		IC-INHERITED, framework-neutral: cyclic-CCC OR R3 single-aeon (choice 'outside scope')
		INHERITED as a LCDM-CDM proxy inside CAMB/CLASS — the standing clusters/CMB debt (see ../dark-sector/)
		INHERITED mechanism — the LCDM-class collisionless account is inherited, not re-derived; relic = native reason a collisionless component is present
		INHERITED at standard-physics-analog level; substrate-graph derivation OPEN
		INHERITED / Higgs-gated; only the RELATIONS among budgets are pinned
		account-tier selection: P09 rules out R (central-scalar), V5/063 composites rule out H
	C SELECTED: P09 scalar phase rules out R; V5/063 composites rule out H	account-tier selection, not a from-scratch derivation
	grounded in GW-00 (D-via-I from saddle geometry in D=3+1)	inherited downstream (Dyn-03/GW-01)
		INHERITED (substrate-c = V1 rate); GW170817 confirms; correctly NOT oversold as ED-distinctive (a shared survival with GR)
		INHERITED (mechanism identified, numbers not recomputed)
		SELECTED (native) + account; asymmetry magnitude inherited

Falsifier / Kill-Cond	Derived	Inherited/Open
		INHERITED (eta_B tier). Mechanism-preferred value is cold GeV-Planck — 6-26 OOM heavier than the needed warm window (the live risk)
		INHERITED (eta_B tier); the relic gives the value a native field-theoretic HOME, amount still inherited
		standing prior result; the sector's strongest, SEPARATE piece (cited not re-derived)
	exact consequence of the combination rule	MOND-SHARED: discriminates ED+MOND from LCDM, NOT ED from MOND; the ED-distinctive part is only the a0 normalization
	real collisionless matter clustering at cluster scale	the warm-keV window keeps it diffuse in galaxies, clumped at clusters — narrow, the live risk
		reconstruction — the Gleason/Piron-Soler chain; orthogonality reduced to operational distinguishability, C selected, Soler lattice rigor the residual
		curvature-emergence, static half closed
	$P = \sqrt{b} e^{\{i p_i\}}$, $Str = \sum P ^2$ splits diagonal (Newton) + off-diagonal (MOND)	standing / structural
		STALE framing: Paper_004 Update 2026-07-02 says 'P-Channel-Orthogonality fully open / three routes fail / blocking'. SUPERSEDED by the reconstruction (next row)

Falsifier / Kill-Cond	Derived	Inherited/Open
		WITHDRAWN 2026-05-17 — 'binary chirality' declared a projection artifact; replaced by R4 single-template + first-arrival IC. Earlier M2 archived in event-density; Memo_ED_BinaryChirality (R1/R2/R3) rendered moot
		RESOLVED / RETRACTED 2026-07-19 -> L_self canonical (two-component). Supersedes the first-draft OPEN flag AND the retracted khronon-mode probe (wrong variable, missed the density response)
		four postulates = 'four facets of one substrate structure' via distinct primitive routes
		above structural analogy, below constructive proof; NOT a Clay-prize-eligible proof (preamble #1,#7)
		genuine ED content = the V5-saturation mechanism only
		arc consolidation
		the reframing row is A->position in its own audit; introduces no new primitives
		A->position ('no platform realizes a horizon'); rests on P-V5-Shared-Mechanism
		A->position — 'does not override or strengthen constituent verdicts'
	form-forced (D-via-I)	higher-order V_n OPEN (not claimed exhaustive)
	form-forced	numerical content canon-internal
	composes closed arcs	carries the Round-4 M1->M2 correction that 095/100 still lag
		NOT a Clay proof; continuum survival OPEN
		five-sector partition not claimed unique; §3.1 RETAINS the stale M1 'FORCED-unconditional' Phase-1 QM label (Staleness #1)

Falsifier / Kill-Cond	Derived	Inherited/Open
		layer-1 native + layer-2 one forced operator + soliton edge (measured negative); full EFE not derived
		the saddle framework subsumes both cosmology papers as one regime
		composition; minimum-tier rule -> M2 base, M3 subdomain carve-outs
		the G-match is an algebraic identity; full nonlinear EFE not from ED's own rule; the 2pi/temperature still via the Euclidean route (open)
	the count-grounding (recognition + attribution; ED already had T19, same count/equipartition/Unruh-inversion/G)	grounds the COUNT, NOT the 1/2-T-per-channel equipartition RULE (T19 posits it, as Padmanabhan does); reaches Newton, NOT Einstein (khronometric wall)
		the triple-identity is stated at SYNTHESIS tier, NOT a theorem (target #15); 4 other singularity PDEs same-shape/unexecuted
		an organizing lens over standing results, NOT a theorem; one rung certified (diffusion)
	the arena + cutoff grounding (via Visser §9)	arena at ACOUSTIC/weak-field tier (curvature emergence OPEN); one-loop R-induction not computed; CC catastrophe DISSOLVED (discrete = no infinite tower; value inherited via H0)
	the equation-of-state derivation (grounds GravityAsEOS §6; discharges its §8 write-out)	the 'fourth face' (scalar-sector co-suppression) KEPT AS CONJECTURE; the 2pi still via the Euclidean route
		the correlation-order invariant is silent on topological defects
		a mnemonic + one reduction (amplitude fusion) + one reclassification; NOT a reduction to two primitives; 3D selected-not-derived
	the real-medium + dynamical-aether grounding; W-W/Marolf obstructions don't bite	kinematic/non-Einstein like the analog field, one bounded step further (weak-field Einstein); Hawking FORCED-CONDITIONAL; generic horizons cold; CC dissolved (value inherited)

Falsifier / Kill-Cond	Derived	Inherited/Open
		capstone; carries a 2026-07-19 correction banner downgrading 'DM is not a particle' to a CANDIDATE + flagging the khronon-alone abundance gap (needs a real relic -> dark-sector)
	argued from the sole-process finding (commitment = the sole process primitive)	NOT forced by a deeper theorem; carries NO delivered prediction yet
	interpretive reframe: Penrose's 'no fact about which time flows' <-> ED's 'one arrow, two branch-clocks, forced to individuate'	quantitative content still Penrose's
		REWRITTEN 2026-09-07; all three clauses were stale. (1) The P14 discharge stands and is the STRONG half, but the row's dependency is now P-Quadratic-Strain, not P14. (2) Paper_030's regime assumption was NOT discharged - it was RESTORED, so 'two postulates to one' reads ONE AND A HALF. (3) The residual is NOT 'constructive sign': that is STRUCTURALLY SETTLED (foundations #10 - $a_r = -d\sigma/dr$, Newton $\sim 1/R$ and MOND $\sim \ln R$ have opposite radial derivatives, so Coh enters with a plus and $\cos \theta_{LH} > 0$ follows). What remains is MEASURED CORROBORATION of canonical Coh, which is candidate because the build measured Grad (#112). Live residuals: threshold generation from a linear horizon response, high-acceleration canonicalization, UFF normalization.
F-COONSET: the two onset at different occupancies; F-BAND: charge & momentum use distinct channels not one shared Adjacency band	co-onset D-via-I (form) given P-Adjacency-Transport-Shared	exact $\eta_{min}:\rho_{max}$ ratio OPEN (needs projection factors Π_p, Π_q)
		form-forced parent; R1/R2 D-conditional; $\rho_{loc}=1$ a structural identity, 0.88 the measured area-law tiling; absolute scale inherited
	the area-SCALING mechanism (direct, vs Bekenstein's indirect BH+second-law)	measured on an ASSUMED geometry (not 'why area from nothing'); grounds horizon-entropy=area + a postulate-conditional channel ceiling, NOT the holographic bound as a general maximum; the 1/4 is thermal (separate)

Falsifier / Kill-Cond	Derived	Inherited/Open
		measured negative: the bare rule disperses (defocusing + dissipative)
		ED's stability route to {1,2,3} tested and REFUTED (symmetric multiplet stable forall N); framework gives SU(N) for any N
		no screw/pseudoscalar at wiring/transport/selection/spontaneous levels
		template reproduced exactly; escape blocked 3 ways; anchored to Sarnak Mobius-disjointness + the sieve parity barrier

Grounded - consistent with, not forced by (re: the budget gives a CONSTANT bound, CKW's is STATE REDUCED, not derived (gravity #48); Paper_097's exp declared in 6 papers (foundations, qm-kinematics, su declared in 6 papers (cosmology, foundations) declared in 5 papers (foundations, qft) declared in 5 papers (cosmology, foundations) declared in 5 papers (foundations, substrate-evaluati declared in 5 papers (foundations, qft) declared in 5 papers (foundations, qft) declared in 5 papers (foundations, qft) declared in 4 papers (foundations, gravity) declared in 4 papers (foundations, predictions, qft) declared in 3 papers (foundations, soft-matter) declared in 3 papers (cosmology, q-compute, soft-ma declared in 3 papers (soft-matter, substrate-evaluatic declared in 3 papers (gravity, qm-kinematics) declared in 2 papers (gravity) declared in 2 papers (foundations, q-compute) declared in 2 papers (foundations, soft-matter) declared in 2 papers (foundations, gravity) declared in 2 papers (foundations, soft-matter) declared in 2 papers (cosmology, soft-matter)

Falsifier / Kill-Cond	Derived	Inherited/Open
		declared in 2 papers (cosmology, soft-matter)
		declared in 2 papers (qft)
		declared in 2 papers (soft-matter)
		declared in 2 papers (cosmology, soft-matter)
		declared in 2 papers (soft-matter, substrate-evaluatic
		declared in 2 papers (qft)
		declared in 2 papers (foundations, predictions)
		declared in 2 papers (qft, relativistic-qm)
		declared in 2 papers (cosmology, soft-matter)
	NOT established. P12 carries no inertial norm	OPEN. ED HAS an inertia mechanism and it is MEASU
	A->regime (the qft arc ledger's own label)	a REGIME CONDITION, stated rather than derived; it
	A->position (the qft arc ledger's own label)	a composite VERDICT over the OS reconstruction, no
	three D-via-I rows; the paper states NO D row	Bloch machinery, the two-scale expansion, effective-

Status	SpotChecked	Postulate
current — the folder's one durable finding		
current		
current		
current		
current		
current		
current — the arc's load-bearing soft spot	yes 2026-07-29	
current — the arc's load-bearing soft spot	yes 2026-07-28	
current		
current		
current — Derived + A->position; NOT claimed to refute W-F mathematically		
current — A->position		
current — A->position		
current — framing, NOT theorems		
current		
current — Asserted (structural, pivot-disclosed)		
current		
current		

Rests On Sign-Critic Last Verified

2026-07-29

2026-07-28

CANDIDATE - confirm, and record the invariant that fixe

2026-07-29

2026-07-29

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
current					2026-09-04
current					
current					2026-07-29
current — 'the beginning carries units, not numbers'	yes 2026-07-29				2026-07-29
current — $G_d = c^3 \cdot I_P^{d-1} / \hbar$ (c-power dimension-BLIND; dimension in I_P^{d-1} =holographic area)	yes 2026-07-29				
current					
current — $\hbar$ = the action-cost of laying down one record (P11)	yes 2026-07-29				
current	yes 2026-07-29				
Current status: superseded — see the correction that follows. Historical correction: STALE AS OF 2026-09-04 — was 'current'. The $1/(2\pi)$ is re-tiered **Postulated/disputed** (gravity Staleness #10): it cancels in Paper_028 §6.3 / Paper_029 §5.1's own displayed algebra. The $\text{SCALE } a_0 \sim cH_0$ stands and is now reached FOUR independent ways (dipole projection; Rindler/de Sitter smoothness; horizon-competition threshold #34; content-depth #38) — none reaches the coefficient. Empirical standing tightened 2026-09-04: Desmond 2023 ($a_0 = 1.19 \pm 0.04 \pm 0.09$) puts the 2π at 1.09 sigma (was 0.49 sigma vs the 2016 systematic), and it PREFERS A HIGH H_0 — 1.50 sigma vs Planck, 0.62 sigma vs SH0ES. See Paper_029 §5.5, Staleness #39					2026-07-29
current					VERIFIED 2026-09-01; 2026-09-07
current					VERIFIED 2026-09-01; 2026-09-07

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
current					2026-07-29
current — under P-Gauge-Orbit-Non-Empty + P-Quotient-Hilbert					
current — no anomaly claim (consistent w/ Lindblad = line-gap, wrong type for anomaly)					
current	yes 2026-07-29				
current					
current					
current — abstract's 'P04-derived' is really postulated (Staleness #3)	yes 2026-07-29		VERIFIED 2026-09-01		2026-09-07
current (the FORM; coefficient has a separate geometric route — next row)					
current					
current					
current			none found - Paper_		2026-07-29
current			Nothing beyond P01		2026-07-29
current — see Staleness #2 on the '2pi derived' claim			Nothing beyond the		2026-07-29
current			Nothing beyond the primitives; the $n_T = -r/8$ consistency relation		
current			P (normalization cho		2026-07-29
current			VERIFIED 2026-09-01		2026-09-07
current — form-IDENTIFIED (M3); P-BinaryAdmission WITHDRAWN; native Z2 relocated to IC not derived (#2b open core persists)	yes 2026-07-29		none found - Paper_		2026-07-29
current — genuine ED-distinctive content			none found - Paper_		2026-07-29

Status	SpotChecked	Postulate	Rests On Sign-Critic Last Verified
current			none found - Paper_ 2026-07-29
current — M3, restricted scope			P-V5-Schmidt-Generic AND P-Bipartite-Mapping. VERIFIED 2026-07-29
current — Derived (conditional)			P-V5-Budget, P-Measure-Saturation; W_max inherited. VERIFIED 2026-07-29
current — Derived (form)			P-Bipartite-Mapping 2026-07-28
current			P-Continuity, P-Addit 2026-09-07
current — M1 (no numerical inheritance); conditional on 017, not primitives-alone	yes 2026-07-29		P-SubstrateLocality. ' 2026-09-07
current — M1 pure dimension-comparison; the cleanest result; a distinctive matter-sector prohibition	yes 2026-07-29		P-NoBackwardChain 2026-07-29
current — Derived (the L2=0 headline is an exact measured zero)	yes 2026-07-29		Nothing beyond a stated position. VERIFIED 2026-09-06 (#125). The row's o
current — quantization measured to machine precision			P-Causality-Propagation. VERIFIED 2026-09-06 (#125). The row's o
current — the in-folder DERIVED route for the 1/4 (imported 2026-07-29)	yes 2026-07-29 (read in full before import)		P-Codim-1, P-Sat; I_ E 2026-09-07
current — Derived (theorem); supersedes The Parity Wall's verdict TIER (not direction)	yes 2026-07-29		P-Potential-Reading 2026-09-07
current — the one genuinely-computable result; 'c^3 = 3D' is a coincidence, dropped	yes 2026-07-29		P-FreeChain, plus the Paper_035 guardrails. VERIFIED 2026-09-06
current			P-MOND-Interpolati 2026-09-04

Status	SpotChecked	Postulate	Rests On Sign-Critic Last Verified
current			P-H0-Cosmological-Id 2026-09-07
current			P-Commitment-Linear. VERIFIED 2026-09-06 (#122): same lapse ch
current			Lovelock uniqueness (conditional) + inherited kappa, Lambda. VER
current — the keystone kill-switch, passed	yes 2026-07-29		none found - Paper_GR-II_KhronometricClass.md declares no cens
current			none found - Paper_GR-II_KhronometricClass.md declares no cens
current	yes 2026-07-29		none found - Paper_GR-II_KhronometricClass.md declares no cens
current — THE one firm upgrade over T17's analogy	yes 2026-07-29		none found - Paper_GR-II_KhronometricClass.md declares no cens
current — Derived-structural			P-Commitment-Linear. VERIFIED 2026-09-06 (#122): the claim text
current	yes 2026-07-29		Nothing beyond P01-P13. VERIFIED 2026-09-06 (#125). C4 FALSE P
current — RESOLVED 2026-07-19 (supersedes the open self-term + the retracted khronon-mode probe)	yes 2026-07-29		none found - Paper_ 2026-07-29
current — the sector's one genuinely-new derived headline	yes 2026-07-29		none found - Paper_GR-IV_ArrowsAlibi.md declares no censused p
current — convention-tagged	yes 2026-07-29		none found - Paper_ 2026-07-29
current — Grounded, NOT pure-M1 (README 'Derived' inflated)	yes 2026-07-29		none found - Paper_054_UR1.md declares no censused postulate

Status	SpotChecked	Postulate	Rests On Sign-Critic Last Verified
current — M3			none found - Paper_054_UR1.md, Paper_056_ClassA_Wall.md de
current — M3 (form); Tsirelson Inherited	yes 2026-07-29		P-Corr-Budget (Pape 2026-07-29
current — M3			Nothing beyond P01-P13, via the QuantumDarwinism accounting 1
current — Grounded/Postulated			P-Polarity-Connection-Match. VERIFIED 2026-09-06 (#125). The pc
current — M3 (identification)			P-Scalar-Match and P-Lorentz-Covariant-Continuum. VERIFIED 202
current — Grounded (conditional)			P-YM-Action-Coarse-Graining. VERIFIED 2026-09-06 (#122): Paper_
current — M2 (Intermediate Path C)			P-Gap-Coercivity, P-F 2026-07-29
current — M3; 1/2-from-primitives Open			Nothing beyond the primitives; fiber-bundle quotient and DCGT cc
current — M2, partly Open			Nothing beyond the primitives; specific L_k and rates inherited. VE
current — inner-product edge Open/SUPERSEDED			none found - Paper_ 2026-07-29
current — M3			P (convention): the s 2026-07-29
current — M3 (lemma + identification)			Nothing beyond the primitives; numerical γ_C inherited. VEI
current — M3; $\hbar$ Inherited	yes 2026-07-29		Nothing beyond the primitives; $e/\hbar$ inherited. VERIFIED 2026-0
current — M3; four-band NOT the archived M-series (clean)	yes 2026-07-29		Nothing beyond the primitives; band content inherited. VERIFIED :
current — Reference-adjacent			none found - Paper_012_7_AdjacencyBandwidth_Galilean.md dec
current — FORM-FORCED under P-Slow-Variation + P-Hadamard-Preservation			none found - Paper_ 2026-07-29

Status	SpotChecked	Postulate
current — conditional on P-Polarity-Connection-Match (postulated)		
current		
current — postulate-construction	yes 2026-07-29	
current		
current — EL step on a postulated action		
current — explicitly NOT a Clay proof	yes 2026-07-29	
current		
current		
current		
current — superseded-in-part by GR-III geodesic treatment (Staleness #6)		
current		

Rests On Sign-Critic Last Verified

none found - Paper_ 2026-07-29

none found - Paper_ 2026-07-29

none found - Paper_ChargeAsTopology_B4.md declares no census

none four CANDIDA1 2026-07-29

Nothing beyond P01-P13; absolute W_max inherited. VERIFIED 20

2026-07-29

2026-07-29

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
current — STRENGTHENED 2026-09-04 (gravity Staleness #37). Paper_030 §5.3 previously reached the deep-MOND branch via an explicit REGIME ASSUMPTION (that the bilocal cross-term dominates two pure-source terms), which was false as stated and load-bearing. Sigma_0 is now a content normalization, no gradient is taken, the profile is $a = a_N + \sqrt{a_N a_0}$, and $a \rightarrow \sqrt{a_N a_0}$ follows as a LIMIT, not an assumption**. Verified numerically over $a_N/a_0 = 1e6..1e-6$. Derived tier better supported than when this row was written. NOTE the mu form is now filed INHERITED (it is the V5 envelope shape, Paper_090 §7.2 value-layer; Staleness #34), not open					
current					2026-07-29
current	yes 2026-07-29				
					2026-07-29
current					2026-07-29
current — weaker than independence					2026-07-29
current — postulate-driven, NOT a proof (README 'proven' overstates, Staleness #4)	yes 2026-07-29				2026-07-29
current — prediction-facing					
current — prediction-facing					2026-07-29

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
current — Derived-structural settlement (2026-07-14)	yes 2026-07-29				
current					
current					
current	yes 2026-07-28				VERIFIED 2026-09-06 2026-07-29
current — strengthens 063 only under that postulate					
current - narrowed 2026-09-04; sheet row updated 2026-09-06					2026-07-29
current — cleanest paper in arc; Bell-Tsirelson explicitly OPEN here (row 11)	yes 2026-07-28				2026-07-28
current — two substrate-derivations open					
current — Derived (conditional)					
current — Derived, M2 (honest non-algebraic exclusion)	yes 2026-07-29				VERIFIED 2026-09-06 2026-07-29
current — Derived (D-via-I)					VERIFIED 2026-09-06 2026-09-07
current — M3 (FORM-FORCED + VALUE-INHERITED)	yes 2026-07-29				
current — M2 (Intermediate Path C); THE #2b gate, genuinely OPEN; chirality NOT claimed derived (vector default, parity inherited). Row-15/§3.7 desync (Staleness #1)	yes 2026-07-29				
current — M3, FORM-INHERITED (correctly reflects 017's round-8 D->P; no over-claim)					

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
current — M3, FORM-INHERITED					
current — M3 (form-IDENTIFIED + VALUE-INHERITED)					
current — M3 (form-IDENTIFIED + VALUE-INHERITED)					
current — M3 (structure forced, existence conditional)					
current — M2 with OPEN flags; NOT built on archived four-band (dwell handle canonically licensed P02+P03+P07)	yes 2026-07-29				2026-07-29
current — M2 (conditional on the P-Gauge postulate); 112's $\sigma=0$ existence depends on this					
current — M3					
current — M3					
current — Grounded/Measured					
current — Grounded (Coulomb open)					
current — resolved grounded-in-kind (not yet a written ED theorem; imported standard einselection)	yes 2026-07-29				
					2026-07-29

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
current — Grounded (structural)					
current — structural result; the distinction is structural, not observational	yes 2026-07-29				
current	yes 2026-07-29				
current — grounded-in-kind scaling argument, NOT a structural derivation; the gap is narrowed, not closed	yes 2026-07-29				
current					
current — 'solid'					
current — reinforces the holographic road; not a 4th witness for 3D					
current — form-IDENTIFIED (M3), value+IC inherited					2026-07-29
current — value inherited via H0, smallness open; do NOT say 'ED inherits the overshoot'	yes 2026-07-29				
current — form-IDENTIFIED (M3), conditional on Route A	yes 2026-07-29				2026-07-29

CANDIDATE - confirm, and record the invariant that fixe

CANDIDA1 2026-07-29

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
OPEN 2026-09-07 - flagship, but the one survey testing it is itself under a coherent systematics challenge; status UNTESTED rather than confirmed-or-refuted. Do NOT over-bank in either direction.	yes 2026-07-29	RE-TIERED Derived -> Grounded 2026-09-07. Previously left at Derived on Paper_029 audit row 14 (evolution independent of the 2pi dispute), which is still true and is not the issue. The issue is a SECOND dependency named in Paper_a0z section 3.6 this session: the exponent also rests on the horizon being read LIVE rather than frozen, which that section calls 'the mechanism's core physical assertion, not a theorem' -- a frozen horizon gives alpha=0 and is indistinguish able from MOND. Workbook Derived			

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
current	yes 2026-07-29				
current — Grounded (observational, order-of-magnitude); MOND cross-link					2026-07-06
current — M3	yes 2026-07-29				
current — M2 (Path C); feeds cosmology Cos-05					
current — M2	yes 2026-07-29				
current — M3					
current — M2 generic (inherits Dyn-03); M3 saturation-case					
current — Grounded (derived-given-inputs; mechanism-tier for the gap)	yes 2026-07-29				
current — Grounded; status = 'reduced to the Soler residual' (NOT blocking/three-routes-fail)	yes 2026-07-29				
current	yes 2026-07-29				2026-07-29
current					2026-07-29
current — M3 (ZERO pure-D rows; 7 D-via-I composition rows)	yes 2026-07-29				2026-07-29
current — D-via-I (form only)					2026-07-29
current — M3					2026-07-29

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
current — M2 net (component composition; inspiral dominates observable bands); M3 subdomain carve-outs	yes 2026-07-29				
current					
CURRENT - standing relativistic MOND embedding; do not reinvent. BUT the high-acceleration canonicalization remains OPEN (2026-09-07): KM-I uses a smooth Cassini-constrained family, Paper_030 a high-acceleration switch; neither is falsified and the canonical choice is unresolved. 'Standing' is the result, not the implementation.	yes 2026-07-29 (corroborated via 030/QuadraticStrain checks)				
current					
current					2026-07-29
current					
current — account; corrected inline (P-imprint retired: C native, P spontaneous+inherited)	yes 2026-07-29				
current — a falsifiable prediction					
current					2026-07-29
current					2026-07-29
current — the topology half is Measured					2026-07-29

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
current — a falsifiable prediction					
current — RESOLVED, near-term weapon closed (consistency result). [CORRECTED 2026-09-04: the banner/body desync was FIXED 2026-07-29 (the paper reads 'Status: RESOLVED' with a pre-resolution note ahead of the 2026-07-21 EFE calculation). What was stale was the ARC-STATE PROSE still listing it as a live exception, corrected 2026-09-04 — gravity Staleness #32. Fourth instance of a staleness note that is itself stale.] superseded text: BODY stale vs its own banner (Staleness #1)	yes 2026-07-29				2026-07-29
current — a genuine foundational advantage over hidden-variable collapse	yes 2026-07-29				2026-09-04
current	yes 2026-07-29				2026-07-29
current	yes 2026-07-29				
current — verdict A->position/M3 under P-Substrate-Cosmology-Unified; ONLY TWO projections (Staleness #2)	yes 2026-07-29				
current - Grounded (gauge-law-forced); sign STRUCTURALLY SETTLED 2026-09-07, not awaiting a measurement	yes 2026-07-29				2026-07-29
current — Grounded (D-structural identity)					2026-07-29
current					
VERIFIED : STRUCTUI 2026-09-07					

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
current					
current					
current — Consilience (M3)					
current — Consilience (M3)					
current — Consilience (M3)					
current — Consilience (M3); F1 falsifier is Route-A4-specific, not a test of slow-roll					
TIGHTENED 2026-09-06 - was 1.7575, then 1.8 +/- 0.3					
current — Measured (on a faithful V5 model)					
current				2026-07-29	
current					
current — explicit consilience, 'not a novel prediction, not tiered as one'	yes 2026-07-29				
current — Measured (tested, came-back-no)					VERIFIED : STRUCTUI 2026-09-07
current — Measured (structural)					

CANDIDATE - confirm, and record the invariant that fixe

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
current — Measured (V5-conditional)					
current — Measured (analytic); form native, smooth field a coarse-grained shadow					
current — toy; supports the §4.2 account only					
current					
CORRECTED 2026-09-06 - read 'P12 coherence rewards alignment but stays FINITE-REACH'; the measured object is Grad 2026-09-07: sign STRUCTURALLY SETTLED; MEASURED CORROBORATION CANDIDATE (same vocabulary as the Report's MOND section).					
CORRECTED 2026-09-06 - claim read '(P12-Coh rewards alignment)' and 'conditional on P12-Coh operationalization' 2026-09-07: sign STRUCTURALLY SETTLED; MEASURED CORROBORATION CANDIDATE (same vocabulary as the Report's MOND section).			VERIFIED : STRUCTUI	2026-09-07	
current	yes 2026-07-29				2026-07-29
open — a proposal/framing note is honest; a 'ED resolves Wallstrom' claim would paper over the unmade identification					
open	yes 2026-07-29				
open					

Status	SpotChecked	Postulate
open		
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open		
open		
open		
open		

Rests On Sign-Critic Last Verified

CANDIDATE - confirm, and record the invariant that fixe

2026-07-29
2026-07-29
2026-07-29
2026-07-29
2026-07-29
2026-07-29
2026-07-29

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
open					
open (§4b, now in-folder)					
open					
open	yes 2026-07-29				
open (#2b = the T4 gate)					
open	yes 2026-07-29				
open	yes 2026-07-29				
open					
open					
open					
open	yes 2026-07-29			2026-07-29	
open					
open	yes 2026-07-29			2026-07-26	
open	yes 2026-07-29				
open					

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
open					
open	yes 2026-07-29				
open — 'reduced to the Soler residual'					2026-07-29
current — 'reduced to the Soler residual': stronger than 004's blocking, weaker than a closed theorem	yes 2026-07-29				
open					2026-07-29
open					
open	yes 2026-07-29				CANDIDATE - confirm, and record the invariant that fixe
open	yes 2026-07-29				
open			VERIFIED 2026-09-06		2026-07-29
open					2026-07-29
open					2026-07-29
open					
					2026-07-29
open	yes 2026-07-29				
					2026-07-29

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
open	yes 2026-07-29				
open					
open					
current — README 'Derived' inflated	yes 2026-07-29	live			
current — foundational commitment		live			
current		discharged			
current		live			
current		live			
current		live			
CORRECTED 2026-09-06 - was Derived, and the claim text read parameter-free	yes 2026-07-29				
current — see QuadraticStrain (Staleness #8)		discharged			
current		live			
current	yes 2026-07-28	live			
current		live			
current		live			
current		live			
current — the whole corpus's 'conditional on Paper_087'	yes 2026-07-29	meta			

2026-07-29

2026-07-28

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
current — matches memory		meta			
current		meta			2026-07-29
current		meta			
current		meta			
current		meta			
current — Postulated (declared, non-primitive)		live			2026-07-29
current					
current		definitional			
current		live			
current — declared postulate		live			
current — the folder's one substantive paper-specific postulate	yes 2026-07-29	live			
current		live			2026-07-29
current — slope-4 + tightness are MOND-SHARED; only the a0 normalization is ED-distinctive	yes 2026-07-29				

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
OPEN 2026-09-07 - flagship, but the one survey testing it is itself under a coherent systematics challenge; status UNTESTED rather than confirmed-or-refuted. Do NOT over-bank in either direction.	yes 2026-07-29				
current — PROVISIONAL (alpha_act, A(x), the discriminator observable all OPEN)					
current — do not upgrade until a substrate tau_V5 floor is supplied	yes 2026-07-29				
current — Prediction (live, shape-level); the folder's genuinely distinctive weapon	yes 2026-07-29				

2026-07-29

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
current — Prediction (weak wedge; NOT distinctive at leading order)					2026-07-29
current — Prediction (domain-restricted-under-tension)	yes 2026-07-29				2026-07-29
current					2026-07-29
current — a sharpened distinctive prediction					2026-07-29
current — a real distinctive null; ties to 22-Ways $w=-1$					
current — CHANNEL-SPECIFIC, NOT a test of inflation itself					2026-07-29
current — directional prediction AND the sector's quantified live risk (mechanism prefers cold)	yes 2026-07-29				2026-09-07
current — clean ED-vs-superfluid-DM discriminator	yes 2026-07-29				

Status	SpotChecked	Postulate
current — ED-distinctive vs MOND	yes 2026-07-29	
current — ED-distinctive lensing	yes 2026-07-29	
current — a real ED-vs-MOND discriminator		
current		
current		
current — passed favorably (alpha_2=0 exact, alpha_1 >=70 orders safe)	yes 2026-07-29	
current — NOT distinctive		
current — knee location rests on a flagged nu~0.70 assumption		

Rests On Sign-Critic Last Verified

2026-07-29

2026-07-29

2026-07-29

2026-07-29

CANDIDATE - confirm, and record the invariant that fixe

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
current — distinctive-in-principle but $\sim 10^{-6}$ -small and unmeasurable near Earth (needs deep-void $g_{ext} < \sim a_0/100$)	yes 2026-07-29				
current — D-conditional on P-QD-LiveWeight					
					2026-07-29
current — the whole corpus is 'conditional on P01-P13'	yes 2026-07-29				
current					
current					
current					2026-07-29
current					
current	yes 2026-07-29				
current					
current					
current					2026-07-29
current			VERIFIED 2026-09-06 (#126). CHECKED, NO CHANGE NEEDED. Led		

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
current — the corpus's keystone primitive	yes 2026-07-29				2026-07-29
current					
current — P13 = time homogeneity, per memory	yes 2026-07-29				
current — scaffolding for 014-023					
					2026-07-29
current					2026-07-29
current — Reference definition + theorems N1/T18					
current — Reference definition (P10-licensed)	yes 2026-07-29				
current — canonical, one stale example	yes 2026-07-29				2026-07-29
current					
current					
current					
current					
current					
current					
current	yes 2026-07-29				
current	yes 2026-07-29				
					2026-07-29
current					2026-07-29

CANDIDATE - confirm, and record the invariant that fixe

Status	SpotChecked	Postulate
current — this route inherits; net 1/4 now derived (Staleness #1 RESOLVED)	yes 2026-07-29	
current		
current		
current		
current		
current	yes 2026-07-29	
current		
current		
current		
current — inherits a Soler residual (see Status/staleness)	yes 2026-07-28	
current		
current		
current		
current — M3 FORM-INHERITED + VALUE-INHERITED		

Rests On Sign-Critic Last Verified

2026-07-29

2026-07-29

2026-07-28

VERIFIED 2026-09-06 (#130). TIGHTENED 2026-09-06: $\xi = 1.740 +$,

2026-07-29

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
current — M3, value-INHERITED with honest circularity	yes 2026-07-29		VERIFIED 2026-09-06	2026-07-29	2026-07-29
current					
current					
current					
current					
current			VERIFIED 2026-09-06	2026-07-29	2026-07-29
current — the #5b debt (distinct from #5c Gaussianity)					
current	yes 2026-07-29				
current					
current					
current			VERIFIED 2026-09-06	2026-07-29	2026-07-29
current					
current					
current					
current					
current	yes 2026-07-29		VERIFIED 2026-09-06	2026-07-29	2026-07-29
current					
current					

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
current	yes 2026-07-29				
current	yes 2026-07-29				2026-07-29
current	yes 2026-07-29 (read in full)				2026-07-29
current — honest distinction (§3.2)	yes 2026-07-29				2026-07-29
current — structural	yes 2026-07-29				2026-09-07
current — consistent with the qm-kinematics/substrate-evaluation Gleason reconstruction	yes 2026-07-29 (read in full)				2026-07-29
current	yes 2026-07-29				2026-07-29
current	yes 2026-07-29				2026-07-29
current — re-checked 2026-09-04 (Staleness #37). The cross-term SURVIVED the Sigma_0 rewrite: Paper_030 §4.2 uses $a_0 R$ as a per-channel CONTENT measure (legitimate; $= c^2/2\pi$ at the horizon), whereas §3.2 took its GRADIENT and called it a force (removed). Same expression, two uses, one licensed. 'Forced' scoped to P14 per the strength-word rule (checklist item 20c)	yes 2026-07-29				2026-07-29
SUPERSEDED - re-tiered from Open 2026-09-06	yes 2026-07-29				2026-07-29
SUPERSEDED by 068 (merged 2026-07-05); kept for provenance					2026-07-05

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
superseded	yes 2026-07-29				
superseded	yes 2026-07-29				
current — Synthesis (M2); explicitly NOT a Hardy/CDP closed-proof; per-theorem closure OPEN					
current	yes 2026-07-29				
current					
current					
current					
current — Synthesis / Asserted					
current (consolidation)					
current — Synthesis				CANDIDA1	2026-07-29
current — Synthesis					2026-07-19
current — Synthesis					2026-09-07
current — Synthesis, honest (Clay-adjacent, correctly tiered)	yes 2026-07-29				
current — Synthesis / framing					

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
current — Synthesis (breaker-passed; per-entry tiers inside)					
current					
current					
current — Synthesis; STALE cross-ref to renamed Khronometric file (Staleness #4)					
current — consilience, not a new derivation					
current — Synthesis					
current — Synthesis					2026-07-29
current					
current — referee-passed; attacks the 'Newton not Einstein' gap from the thermodynamic side					2026-07-29
current — Synthesis / Grounded					2026-07-29
current — Synthesis / Grounded					
current					

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
current					
current — structural synthesis, unconfirmed (Report §6 guardrail carried verbatim)	yes 2026-07-29				
current — the reframe, not a new number	yes 2026-07-29				
current 2026-09-07 - STRUCTURAL, conditional on P-Quadratic-Strain; regime assumption restored; sign structurally settled, measured corroboration candidate	yes 2026-07-29				
current — Synthesis (form-forced parent)					
current — consilience, not a novel prediction					

2026-07-29

CANDIDATE - confirm, and record the invariant that fixe

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
current — Wall (measured edge); not 'solitons impossible' (needs an added focusing medium)					CANDIDA1 2026-07-29
current — Wall (#2b); unsolved in standard physics too; must NOT be tiered closed	yes 2026-07-29				
current — Wall (#2b); superseded in TIER (not direction) by CleanSubstrateVector; P inherited					
current — Wall (proven); the program's ONLY theorem-anchored proven negative	yes 2026-07-29				
current		P-V5-Budget,	VERIFIED	2026-09-06	2026-09-06
current - split off row 340 on 2026-09-06			VERIFIED	2026-09-06	2026-09-06
added 2026-09-06 (C1 coverage)		declared; live	VERIFIED	2026-09-06	2026-09-06
added 2026-09-06 (C1 coverage)		declared; live	VERIFIED	2026-09-06	2026-09-06
added 2026-09-06 (C1 coverage)		declared; live	VERIFIED	2026-09-06	2026-09-06
added 2026-09-06 (C1 coverage)		declared; live	VERIFIED	2026-09-06	2026-09-06
added 2026-09-06 (C1 coverage)		live	VERIFIED	2026-09-06	2026-09-06
added 2026-09-06 (C1 coverage)		declared; live	VERIFIED	2026-09-06	2026-09-06
added 2026-09-06 (C1 coverage)		declared; live	VERIFIED	2026-09-06	2026-09-06
added 2026-09-06 (C1 coverage)		declared; live	VERIFIED	2026-09-06	2026-09-06
added 2026-09-06 (C1 coverage)		declared; live	VERIFIED	2026-09-06	2026-09-06
added 2026-09-06 (C1 coverage)		live	VERIFIED	2026-09-06	2026-09-06
added 2026-09-06 (C1 coverage)		declared; live	VERIFIED	2026-09-06	2026-09-06
added 2026-09-06 (C1 coverage)		declared; live	VERIFIED	2026-09-06	2026-09-06
added 2026-09-06 (C1 coverage)		declared; live	VERIFIED	2026-09-06	2026-09-06
added 2026-09-06 (C1 coverage)		declared; live	VERIFIED	2026-09-06	2026-09-06
added 2026-09-06 (C1 coverage)		declared; live	VERIFIED	2026-09-06	2026-09-06
added 2026-09-06 (C1 coverage)		declared; live	VERIFIED	2026-09-06	2026-09-06
added 2026-09-06 (C1 coverage)		live	VERIFIED	2026-09-06	2026-09-06
added 2026-09-06 (C1 coverage)		declared; live	VERIFIED	2026-09-06	2026-09-06

Status	SpotChecked	Postulate	Rests On	Sign-Critic	Last Verified
added 2026-09-06 (C1 coverage)		declared; live	VERIFIED 2026-09-06		2026-09-06
added 2026-09-06 (C1 coverage)		declared; live	VERIFIED 2026-09-06		2026-09-06
added 2026-09-06 (C1 coverage)		declared; live	VERIFIED 2026-09-06		2026-09-06
added 2026-09-06 (C1 coverage)		declared; live	VERIFIED 2026-09-06		2026-09-06
added 2026-09-06 (C1 coverage)		declared; live	VERIFIED 2026-09-06		2026-09-06
added 2026-09-06 (C1 coverage)		declared; live	VERIFIED 2026-09-06		2026-09-06
added 2026-09-06 (C1 coverage)		declared; live	VERIFIED 2026-09-06		2026-09-06
added 2026-09-06 (C1 coverage)		declared; live	VERIFIED 2026-09-06		2026-09-06
added 2026-09-06 (C1 coverage)		declared; live	VERIFIED 2026-09-06		2026-09-06
open - added 2026-09-06; upstream of the weak-field classical tests			VERIFIED 2026-09-06		2026-09-06
current - scope condition on the Yang-Mills rows (019/021/022)			VERIFIED 2026-09-06		2026-09-06
current			VERIFIED 2026-09-06		2026-09-06
current - M2 (Intermediate Path C)			VERIFIED 2026-09-06		2026-09-06

s the sign

inherited. VERIFIED 2026-09-06 (#125). The row's own cell labels the cross-arc identification D-via-I.

9-06 (#124), read against the paper's own audit table. Paper_064 audit rows 2 and 8 are both labelled P; the D row (rank>1 => non-factorizable) sits on top of

026-09-06 (#124), read against the paper's own audit table. STALE ROW: the claim previously read 'Monogamy (CKW-form inequality)'. Paper_065 and the en

ie row's own Derived cell reads literally 'position' - the paper's A-position label - while the Tier column read Derived. Self-contradicting on its face, exactly like
wn cell read: inheritance postulated (P-Causality-Propagation); specific arrow derivations OPEN.

(#125). The row's own cell read FORCED-CONDITIONAL on P-FreeChain, flagged the eikonal-to-action step OPEN, and recorded verdict M3, the weakest of the

tain as row 63. [re-tiered Derived -> Grounded, 2026-09-06, gravity ledger #122]

IFIED 2026-09-06 (#122): GR-II preamble 2 makes the form conditional; preamble 7 -- ' $\kappa = 8 \pi G$ and Λ are inherited, not derived'. Form derived, v

used postulate (mechanical check only, 2026-09-06; resolved EXACT)

used postulate (mechanical check only, 2026-09-06; resolved EXACT)

used postulate (mechanical check only, 2026-09-06; resolved EXACT)

itself names the postulate. Self-contradicting as filed. [re-tiered Derived -> Postulated, 2026-09-06, gravity ledger #122]

OSITIVE: GR-III declares P-Commitment-Linear, but that postulate carries row 69 ($\alpha=1$), not this claim. GR-III's audit labels $c_s = c$ a plain D (leading order)

postulate (mechanical check only, 2026-09-06; resolved EXACT)

(mechanical check only, 2026-09-06; resolved EXACT)

declares no censused postulate (mechanical check only, 2026-09-06; resolved ALIAS)

theorem (sheet row 95, which also stands). VERIFIED 2026-09-06 (#125). C4 FALSE POSITIVE: Paper_058b declares P-Corr-Budget, which carries row 76, not the postulate was sitting in the row's own Derived cell (DCGT coarse-graining + P-Polarity-Connection-Match), not in the Postulate column.

6-09-06 (#125). Paper_017's audit labels this step D, but the object it is about, the continuum free-scalar equation, is built by that same audit as a P (definition _019 section 3.4 -- 'standard variational derivation from the P-postulated action ... the action itself is constructed via P-YM-Action-Coarse-Graining'. [re-tiered

imposed. VERIFIED 2026-09-06 (#125). The row's own cell reads D-via-I. Constructive Gribov resolution remains OPEN.

VERIFIED 2026-09-06 (#125). The row's own cell reads FORM-FORCED as the Markovian limit (D-via-I). Non-Markovian OPEN.

VERIFIED 2026-09-06 (#125). The row's own cell reads D-via-I. UPSTREAM CAVEAT: the holonomy is taken on Paper_016's connection, which row 78 now shows I

9-06 (#125). The row's own cell reads D-via-I, and the paper states plainly that ED does NOT newly predict AB. Same upstream caveat as row 86.

2026-09-06 (#125). The row's own cell reads D-via-I (abelian rep theory).

declares no censused postulate (mechanical check only, 2026-09-06; resolved ALIAS)

ed postulate (mechanical check only, 2026-09-06; resolved ALIAS)

26-09-06 (#125). C4 FALSE POSITIVE: the paper declares P-QD-LiveWeight, but its audit attaches that to the GHZ-width ceiling, a different claim. The accountir

s the sign

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ger #112 corrected the ATTRIBUTION of P12's coherence term, not the formula: 'The formula stands; the attribution does not.' $\text{Sigma} = \text{Coh} - \text{Str} - \text{Grad}$ is uncl

s the sign

± 0.028 lu (40 seeds, 1600 snapshots, same estimator imported so the method cannot drift). The ± 0.30 previously here is the per-SNAPSHOT spread, not th

s the sign

them and the paper's own conclus

tanglement arc ledger BOTH narrow

row 69 was.

: three. Arguably weaker than Grou

alues inherited. [re-tiered Derived

, with the reserve-dissipative step :

is claim.

nal construction) under P-Scalar-M

Derived -> Grounded, 2026-09-06,

rests on P-Polarity-Connection-Mat

ing theorem and the unbounded-re

anged. What moved is that the bu

e uncertainty on the mean: per-snr

